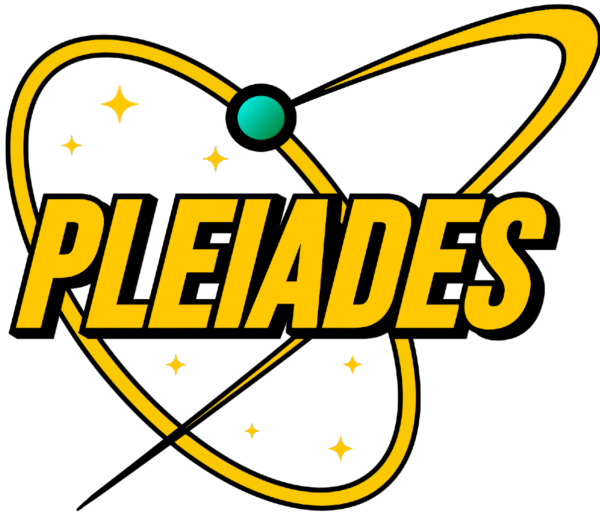




PLEIADES

Payload Launch & Experimental Instrumentation for Advanced Deployment & Exploration System

Concept Design Report for PLEIADES with Payload Deployment

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Executive Summary

The senior design team consists of three AE students, one ME student, two EE students, and one CE student, bringing together an interdisciplinary skill set to support the successful deployment of this rocket mission. The rocket developed for this project is named PLEIADES, inspired by the star cluster representing a family of seven stars. The name reflects both the seven members on the team and the unity required to bring this interdisciplinary project to life.

This senior design project focuses on the development of a high-power rocket capable of deploying two independent payloads sequentially after apogee during descent, followed by an additional ground payload after landing. To support this functionality, the ECE students will be developing a custom avionics system to enable sequential, recoverable payload deployments at pre-determined altitudes as well as collect and transmit real-time flight data to a ground station for monitoring pre-flight, flight, and post-flight analysis. In the aerospace industry, many companies face financial constraints during missions since most launch systems deploy only one payload per flight. This limitation can create barriers for advanced experimentation of deployment mechanisms, raise operational costs, and can reduce accessibility to small companies and commercial users. This project aims to address these challenges by having multiple payload deployments in a single flight, allowing each payload to be released at different altitudes for specific mission objectives.

The deployment system integrates dual deployment mechanisms, redundant electronics, and independent recovery systems to ensure safe, reliable, and controlled payload release without compromising the stability or recovery performance for the ground payload deployment. Some

benefits of this project application include lower launch costs, increased flexibility, and open up experimentation to support multi-stage testing in rapid succession.

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Terms and Abbreviations

- 1) QFD: Quality Function Deployment
- 2) FMEA: Failure Modes and Effects Analysis
- 3) MCU: Microcontroller Unit
- 4) IMU: Inertial Measurement Unit
- 5) PCB: Printed Circuit Board
- 6) RF: Radio Frequency
- 7) GPS: Global Positioning System
- 8) GNSS: Global Navigation Satellite System
- 9) USB: Universal Serial Bus
- 10) SPD: Sequential Payload Deployment
- 11) DC: Direct current
- 12) ADJ: Adjustable
- 13) EMI: Electromagnetic interference
- 14) SMA: SubMiniature version A
- 15) M1: Milestone 1
- 16) M2: Milestone 2
- 17) M3: Milestone 3
- 18) M4: Milestone 4
- 19) M5: Milestone 5
- 20) M6: Milestone 6
- 21) SM: Stability Margin
- 22) PLA: Polylactic Acid
- 23) FPGA: Field-Programmable Gate Arrays
- 24) PWM: Pulse Width Modulation
- 25) EDA: Electronic Design Automation
- 26) SMA: Sub-Miniature Version A
- 27) VCC: Voltage Common Collector
- 28) SPI: Serial Peripheral Interface
- 29) LoRa: Long Range
- 30) LEO: Low Earth Orbit
- 31) m: Mass
- 32) I_{motor} : Motor Impulse
- 33) I_{grav} : Gravity Impulse
- 34) L_{tot} : Total Impulse
- 35) h_b : Burnout Height
- 36) v_b : Burnout Velocity
- 37) Δh : Kinematic Height
- 38) h_{tot} : Apogee Height
- 39) D: Drag Force
- 40) a: Average Acceleration
- 41) RPN: Risk Priority Number

I. Introduction

The purpose of this report is to present the development of the PLEIADES rocket mission for senior design and to document the progress achieved during the current phase of this project. This report provides a continuation of the project background and user needs, defines engineering requirements, and summarizes the relevant technology research that supports the implementation of the separate systems (the vehicle, avionics, and payload deployment mechanisms). It also discusses the concept generation process and initial design implementations, along with preliminary engineering analyses and early vehicle motor testing. Additionally, the report outlines the projected progression of the project for the upcoming semester through a detailed Gantt chart, evaluates potential failure modes through a Failure Modes and Effects Analysis (FMEA), and concludes with a discussion of current work and future tasks required to complete the PLEIADES mission.

High-power rocketry has grown substantially over the years across academic, commercial, and hobbyist sectors due to decreasing material costs, advancements in technologies like additive manufacturing, and continued interest in small-scale aerospace applications. University organizations typically rely on high-power rockets to test experimental avionics, payloads, aerodynamic configurations, and recovery systems. However, commercial applications are on a much larger scale compared to high-power model rocketry. In these applications, deploying a single payload during one mission can create inefficiencies as well as increased costs.

The problem addressed by this project is the lack of an efficient, reliable method for deploying multiple payloads sequentially during a single flight. Creating a rocket capable of

sequential multi-payload deployment has the potential to increase flexibility in a mission while also reducing financial and scheduling barriers associated with multiple launches.

The motivation of this project is the chance to introduce a cost-effective solution that enables multiple payloads to be launched reliably and analyzed in a single flight. This project also provides value through the creation of a custom avionic and telemetry system that can track all flight data in order to analyze the performance of the flight and payload mechanism deployments. The data that is collected can improve the reliability of the deployment mechanisms and can provide additional information that can be evaluated to further improve the designs.

To address this problem, the PLEIADES team decided to only release payloads after apogee was reached. There will be two deployment mechanisms in order to deploy two separate payloads during descent safely. Once the rocket reaches the ground there is an additional payload that will be released on the ground.

The remainder of this report is separated to document the full development process of the PLEIADES mission. First, the customer needs and engineering requirements are defined and explained, followed by a summary of the relevant background research. Then, the concept generation and preliminary analyses are elaborated on, along with the schedule and development plan for the next semester. A FMEA evaluation is included to address the potential failure modes and ways to mitigate these potential risks. The report concludes with a summary of our current progress and identifies future tasks required to complete the project.

II. Needs Analysis

Designing a successful engineering system requires a clear understanding of who the users are and what problems they need solved. For this project, user needs were collected from team discussions, advisor meetings, comparisons to similar rocketry systems, and operational constraints typical of high-power rocket missions. These needs were then organized using a functional and component decomposition, and prioritized to guide technical specification development and ensure that the rocket meets the expectations of all stakeholders and potential users.

The end users of this system are the rocket design team, who are responsible for assembling the rocket, loading payloads, configuring avionics, and recovering the payloads and rocket at the end of the demonstration. The design must be stable, safe, easy to assemble and reliable so that the system performs consistently during testing and the final demo. Stakeholders include the advisors who will review the final project and customers who need a system to deliver their payloads. To satisfy the advisors, the rocket must achieve its goals as stated in the proposal, adhere to engineering and safety standards, and demonstrate a clear traceable design process. To satisfy the customers, the payloads must be safely deployed, delivered on time, and be easily recoverable. Safety, ease of operation, modularity, and size constraints are also a major concern.

The highest priority user requirement for all groups is to ensure a safe and stable flight trajectory throughout the entire mission. Launch operators and faculty reviewers expect the rocket to perform consistently without uncontrolled motion or instability. Users also require strong aerodynamic efficiency and portability, as good aerodynamics improve altitude performance. This drives engineering decisions regarding structural integrity, mass distribution,

aerodynamic layout, and reliable avionics. Meeting this need ensures the safety of the people at the launch site, the test environment, and the payloads.

A key requirement for end users and stakeholders is the ability to carry and deploy multiple payloads at predetermined altitudes. Users expect the deployment mechanism to function reliably and without interfering with other subsystems. This requirement influences the payload bay layout, deployment mechanism, sensor selection, and power management strategy. Proper deployment timing also depends heavily on accurate telemetry and sensor data, linking avionics performance directly to mission success.

Affordability is a core need expressed by purchasers and educational stakeholders. The system must deliver the required performance while remaining cost-effective and maintainable. Lightweight and modular construction facilitates transportation, assembly, and last-minute field adjustments. This need guides decisions in material selection, manufacturing processes, component sourcing, and the design of modular PCBs that can be easily repaired or replaced.

Another key factor for our senior design project is real-time and post-flight data acquisition at the ground station. Users rely on telemetry to monitor acceleration, velocity, altitude, orientation, and position. Additionally, onboard data storage ensures mission data is preserved even if communication is interrupted mid-flight. Accurate position and altitude tracking are especially important for payload release coordination and efficient post-flight recovery. These needs directly dictate requirements for the telemetry system, antenna selection, ground station, and data handling architecture.

Finally, operational simplicity is an important user need. Launch operators require the rocket to be easy to assemble, inspect, and modify, particularly during pre-launch field

operations. This drives a modular approach to design, enabling quick repairs and streamlined pre-launch workflows.

Integrating the needs of all identified user groups—launch operators, faculty stakeholders, and future users — provides a comprehensive foundation for system requirements. Safety, rocket functionality, data reliability, cost-effectiveness, and ease of operation emerged as the core priorities. By understanding the users and tailoring the system to their needs, the engineering team ensures that design decisions align with what matters most, ultimately supporting a successful and reliable mission. The full list of user needs are presented in Appendix B - Table 15.

III. Technology Assessment

Aerostructures

When selecting the material for the airframe, the group researched several available options including kraft cardboard, phenolic cardboard, blue tube, fiberglass, and carbon fiber. After weighing performance, cost, structural requirements, and manufacturability, the team decided on choosing from one of the cardboard material types for the airframe. This decision narrows the options to the kraft cardboard, phenolic cardboard, and Blue Tube as the potential airframe material.

Kraft Carboard

Kraft cardboard tubes are the standard selection for model rocketry due to the tubes being the most affordable airframe material. This material is beneficial for achieving higher altitudes and better thrust to weight ratios due to the material being lightweight. The material is also very easy to modify for assembly and compatible with other components such as the plywood for the

bulkheads, centering rings, and fins. However, kraft cardboard has the lowest compressive strength which can possibly buckle at high thrust loads. It is also susceptible to moisture which creates sensitivity and can warp with humidity if not properly stored. In the unfortunate case that the shock cord deploys rather aggressively, the material can possibly zipper (tear or cut in the body tube) from the strong forces. All in all, this material is still a great option when there is a good amount of thrust and good reinforcement with materials like epoxy.

Phenolic Cardboard

Phenolic cardboard is similar to the kraft cardboard with the addition of additive resin for additional compressive strength and superior thermal resistance near the motor. However, this material tends to be more brittle and can crack upon impact so it has very little elasticity. This material also costs more than plain kraft cardboard. It can also be more difficult to machine if there needs to be sanding and cutting required.

Blue Tube

The last material to discuss for the airframe is Blue Tube. This material is a resin cellulose composite fiber that was engineered specifically for rocketry. It offers very high durability and can withstand more impact than both the phenolic and kraft cardboards. This material is also moisture resistant, so is not easily affected by the weather conditions. Since it is moisture resistant, it is suitable for outdoor conditions and repeated flights. Blue tube has high compressive strength and usually does not require additional reinforcement when compared to the kraft cardboard. However, this material is more dense than the other two materials and is more expensive than the kraft cardboard. It is also the most difficult of the three to machine and integrate by requiring more tooling for clean cuts.

Plywood

The optimal material for the centering rings, bulkheads, and fins is balsa plywood. The most advantageous quality is that it combines the strength of the plywood while also being lightweight. It is rather simple to work with and can be easily manufactured (shaped, cut, and sanded) with simple and accessible tools. Also, the plywood is very versatile, being able to adjust the thickness of selected components if needed. Plywood components integrate effectively with cardboard airframes and provide stable mounting surfaces for structural and recovery hardware.

PLA Plastic

For the nose cone, PLA plastic was selected due to its low cost, versatility, and compatibility with iterative manufacturing. 3D printing allows the group to precisely decide and control the geometry, dimensions, and features of the rocket, and allows for rapid iteration advancements during the designing stage in the project.

Propulsion

Propulsion is likely the most distinctive aspect of rocketry. The propulsion system is responsible for adequately launching the rocket and hopefully reaching some specified altitude. In the case of this project, the altitude goals revolve around giving the vehicle enough time to deploy all the payloads that are intended to be deployed during descent before the rocket lands. Most high-powered rockets built within a modest budget and require simplicity lean towards the selection of solid rocket motors instead of liquid engines. Furthermore, the team is constrained under the Level 1 motor impulse classifications, landing solidly on the I-class motor as the most powerful motor the team can use. Given the unique deployment system this vehicle will be using

(and its added weight), the maximum impulse selection of this class of motor will be the intended powerplant of the rocket (around 640 Ns).

The current top candidates in motor selection are AeroTech I500-14A, AeroTech I280DM-14A, AeroTech I65-10, and AeroTech I132-14. AeroTech is a relatively easy brand to attain motors from and these specific selections showcase a promising amount of power. Additionally, to reduce the complexity of the recovery system, motors that weren't plugged were prioritized, leaving the team with the given list.

Rocket Recovery System Design

Recovery is a critical part of any high-powered rocket, especially one that will not reach outer space. The purpose is to ensure the safe return of the rocket, its payload, and any avionics systems while also preventing damage to surroundings on landing. Most rockets use a parachute-based system due to its reliability, relatively light weight, and simplicity to install. In a dual-deployment system, which we have elected to use, a drogue parachute is first deployed at apogee to initially slow the descent of the rocket, followed by a later deployment of the main parachute to ensure a safe touchdown velocity. Parachute selection and sizing requires a balance, as the chute must be large enough to slow down the rocket for touchdown, but not too big such that it is adversely affected by wind conditions.

The ideal form of main chute for our rocket is going to be a hexagonal shape. This provides plenty of surface area to slow down the rocket while also maintaining some redundancy. Since there are six different cables attached from the chute to the rocket, one at each corner, a failure in one of them will not cause a detrimental failure to the rocket. Speed may not decrease as drastically as expected, but it will still come to a reasonable level before landing.

These parachute designs are also easy to pack for launch, as they do not have as many loose cables as designs such as a hemispherical chute or cruciform, which can have more than double the amount. As we will be placing these on any loose components, including ejected payloads, easier loading is definitely a benefit.

When looking at our drogue chute, it is important to remember the overall reason for using one. A drogue is intended to gradually reduce the speed of the rocket, not slow it down to landing speeds. Since the rocket will be so far in the air, we want it to fall mostly under its own weight, so that the wind does not take over and blow it miles away. The drogue lets this happen while still reducing speed enough so that the main chute deployment is not overstressed by the impulse of energy from catching the air. The design that we intend to use here is a ribbon chute, which provides ample drag and stability to its descent while not excessively slowing down the rocket. Deploying this drogue at apogee will be facilitated by a split in the rocket body. The drogue will slow the descent slightly so that the split body can be used to deploy payloads, then the main chute will be opened closer to the ground for landing.

To deploy a parachute, of course some form of ejection system is needed. After looking at CO₂ and pyrotechnic charge-based systems, the ideal system here is one that uses a pyro charge. These use black powder to create a small burst, which will pop open the main body of the rocket and eject the parachute from within it. There are two different types of pyro charges in model rocketry, and we may be using both. The first is tied to the motor. At the completion of its main burn, a secondary charge will light and ignite upwards into the body instead of downwards, which forces open the body. This would be ideal for our drogue chute, as it would be the first deployment, however the drogue can also be deployed separately by using an IMU that would detect when the rocket is no longer moving upwards. Either can be used effectively, but as we

want to maximize the altitude we achieve, a separate charge is likely better. When using a separate charge, these can either be timed and triggered by the electronics system, or can be tied to a sensor like the IMU. These can be used to accurately deploy a chute exactly when needed, which will be useful for not just the drogue, but the main chute too. By using the IMU and some simple calculations of how quick our parachute can slow down the rocket, we can determine the absolute minimum the chute must be deployed by for the rocket to land safely.

When looking at some basic calculations for drag, we can create a good estimation for how large our parachute will need to be for both the main chute and the drogue. First, we can look at the drogue chute. These should slow the rocket down to the range of 18-25m/s. While this is still way too fast for landing, it ensures that the rocket is not speeding up on descent, and provides a safe environment for our payload deployments. The drag balance equation can give us a good estimate of our area here:

$$A = \frac{2 * m * g}{\rho * v^2 * C_d}, \text{ where } m \text{ is mass, } g \text{ gravity, } \rho \text{ air density, } v \text{ target speed, and } C_d \text{ drag.}$$

By comparing the assumed drag coefficients of the canopy, which for a ribbon chute is approximately 0.50, with an assumed mass of roughly 10kg, we can determine that our drogue chute should be roughly 0.50m². For the main chute, the intended descent speed should be near 8-10m/s to ensure a safe and uneventful landing. With an assumed drag coefficient of 0.80 for the hexagonal chute, we can again determine that our main chute should be roughly 2.5m². These chutes, when deployed at their respective altitudes, will ensure that our rocket slows down enough to safely perform deployments and landings without being blown away by the wind.

Telemetry & Communication Systems Design

Reliable telemetry is one of the most critical elements of the SPD rocket's avionics architecture, as it provides real-time flight data to the ground station and supports post-flight analysis. The telemetry system must operate over long distances, maintain signal integrity during high-speed ascent, and handle intermittent line-of-sight loss while minimizing power consumption. Several wireless communication technologies were evaluated to determine the most practical solution for transmitting altitude, GPS coordinates, orientation data, and general system status. The telemetry subsystem is responsible for transmitting live flight data—including GPS coordinates, altitude, velocity, and mission status—from the airborne avionics' unit to the ground station. This data is then interpreted by our ground-station software and displayed on a laptop-based visualization platform (Google Maps interface).

Acquiring the proper transceiver is necessary for the design of our telemetry system as it is responsible for sending the data to the ground station and receiving it to display on our laptop. We decided to make a tradeoff matrix between different Long Range RF modules to see which best fits the design for our rocket. We chose between RFM95W-9152S, E22-900M30S, XBee-PRO, and HC-12. After evaluating performance, integration complexity, power consumption, RF capability, and compliance with project constraints, the RFM95W-915S2 was selected as the transceiver for the telemetry subsystem. It provides an excellent balance between range and efficiency, offering sufficient link budget to maintain communication throughout the rocket's flight while requiring far less current than the high-power E22-900M30S. Its compact form factor and surface-mount footprint support clean integration onto the custom PCB, and the SPI interface is fully compatible with the ESP32 microcontroller.

We must select the proper antenna for the telemetry subsystem with our options being; Patch antennas, PCB antennas, and simple dipole antennas were compared based on gain, size, weight, orientation sensitivity, and manufacturing complexity. Patch antennas offer stable gain and directional performance but may be too bulky for certain rocket configurations. PCB antennas are lightweight and integrate cleanly into the avionics boards, though they tend to have lower gain. Dipole antennas provide a balanced compromise of size and performance and are commonly used in LoRa-based telemetry systems. Ultimately, antenna selection must consider not only RF performance but also mechanical stability and the constraints of the electronics bay.

After evaluating the gain characteristics, form factor, mechanical constraints, and RF behavior of each antenna category, the telemetry subsystem will use a passive quarter-wave whip antenna at 915 MHz for the LoRa transceiver. This type of antenna provides the most reliable omnidirectional radiation pattern, ensuring stable communication regardless of the rocket's attitude or roll rate. Meanwhile, the GPS subsystem will employ an active patch antenna, which is optimized for receiving low-power GNSS signals and provides improved sensitivity and tracking performance throughout the flight. This dual-antenna approach allows the telemetry and navigation systems to operate independently while maximizing signal reliability, range, and system robustness during high-altitude flight.

The final telemetry architecture with the RFM95W-915S2 transceiver module and 915 MHz whip antenna and Taoglas active GPS patch antenna was chosen to satisfy range requirements, minimize power consumption, and maintain reliable communication across the full flight envelope. LoRa provides the necessary link of robustness and distance capability while keeping the hardware lightweight and energy efficient. Combined with the ESP32's processing capabilities and native USB support for the ground station, this communication system offers a

practical, cost-effective solution for collecting and delivering real-time flight data during the SPD mission. The LoRa whip ensures long-range omnidirectional communication throughout ascent and descent, while the active patch antenna provides precise navigation input to support real-time tracking and post-flight recovery. These models integrate cleanly into the final PCB through standard RF connectors and provide the necessary electrical performance for reliable high-altitude flight operations.

Microcontroller Unit Selection

Several processing technologies were evaluated, beginning with a comparison between Field-Programmable Gate Arrays (FPGAs) and microcontroller units (MCUs). For our system, many aspects of FPGAs would be unnecessary and require more complexity for the circuit. For the system, we will use MCUs as they are cheaper, have lower power consumption, are easier to use, and have broader implementation. To make sure we choose the appropriate MCU for our system, we explored various options for their architecture performance, power consumption, communication peripherals, and cost. After evaluating all platforms, the ESP32 was selected as the most suitable microcontroller for the SPD rocket. Its combination of processing speed, peripheral availability, power efficiency, integrated wireless communication, open-source development support, and cost effectiveness provides the best balance for a mission requiring continuous sensor polling, real-time decision making, and long-range telemetry. To finalize the choice of ESP32 model, three variants, ESP32-WROOM-32E, ESP32-S3-WROOM-1, and ESP32-C3-MINI-1, were compared using a trade-off matrix that evaluated processor capability, available interfaces, operating voltage, power consumption, physical size, cost, and appropriateness for the rocket's avionics layout. This comparison ensures that the selected module aligns with both the electrical and mechanical constraints of the system.

Based on the trade-off matrix presented, the ESP32-S3-WROOM-1 was selected as the final microcontroller for the SPD avionics system. Among the three evaluated ESP32 variants, the S3-WROOM-1 provides the strongest overall balance of processing capability, power efficiency, available interfaces, and physical size.

Sensor Technology Selection

The sensor suite for the SPD rocket was selected to provide accurate, reliable, and redundant measurement of flight conditions necessary for safe payload deployment and post-flight recovery. After evaluating multiple options for altimetry, inertial measurement, global positioning, and acoustic recovery, the final configuration was chosen to balance precision, power efficiency, size constraints, and integration simplicity within the custom avionics system.

For altitude measurement, the MS5611 barometric pressure sensor was selected due to its exceptionally high resolution, low noise, and rapid response time, making it suitable for detecting apogee and triggering altitude-based deployment events. Its performance exceeds that of lower-resolution alternatives such as the BMP280 and BMP390, providing the stability needed for precise altitude tracking in a dynamic flight environment.

For inertial sensing, the ICM-20948 9DoF IMU was chosen because it offers accurate accelerometer, gyroscope, and magnetometer measurements in a compact, low-power package. Its advanced filtering, low drift characteristics, and full 9-degree-of-freedom capability outperform older 6DoF modules, ensuring that the system can reliably monitor orientation and confirm stable flight conditions prior to payload release.

For positional data used during descent and recovery, the u-blox MAX-M10S GPS module was selected. While lower-dynamic than high-end receivers such as the NEO-M9N or

ZED-F9P, the MAX-M10S meets mission requirements because GPS is not used for real-time flight control. Instead, it requires a satellite connection during coast and descent, providing accurate landing coordinates with minimal power consumption and small form factor.

Finally, for acoustic recovery, a passive piezo buzzer driven through a MOSFET was selected. This option provides a lightweight, power-efficient method for generating a distinct audible tone to assist recovery teams in visually obscured terrain. Active buzzers and electronic beeper modules were evaluated but found to be heavier, more power-intensive, or unnecessarily complex for the project's needs.

METRIC	Passive Piezo Buzzer	Active Buzzer	Electronic Beeper Module
Sound Level	~70 dB	~90 dB	95 dB+
Power Consumption	10 mA	20 mA	25 mA
Control	Needs PWM signal	Simple ON/OFF (self-oscillating)	Simple ON/OFF
Size	Very small	Small	Medium
Cost	Very low	Low	Moderate

Table 01. Comparison of Beepers

Together, the MS5611, ICM-20948, MAX-M10S, and passive buzzer form a cohesive sensor suite that meets all mission performance, power, and integration requirements. This combination provides high-quality data for flight monitoring, supports reliable payload deployment, and ensures successful post-flight recovery while minimizing system complexity and mass.

Actuator Selection

The payload deployment system requires actuators that are mechanically reliable, electrically compatible, and capable of producing controlled motion under the dynamic conditions of rocket flight. The type of actuator selected directly influences the mechanism design, response speed, torque output, and overall power budget. Several actuator types were evaluated for use in the SPD rocket, each offering distinct advantages and limitations depending on the required motion—rotary, linear, or binary release.

Actuator	Motion Type	Precision	Energy Efficiency	Response Speed	Control Complexity
Servo Motor	Limited-angle rotary	High	Moderate	Moderate	Low (PWM)
Solenoid	Linear push/pull	Low	Low	High	Very Low
Linear Actuator	Linear, controlled stroke	Very High	Low	High	Medium
Stepper Motor	Incremental rotary	High	Moderate–Low	Moderate	Medium
DC Motor	Continuous rotary	Low–moderate	High	Variable	Low

Table 02. Comparison of General Actuators

After evaluating servo motors, solenoids, linear actuators, stepper motors, and DC motors, a micro servo motor was selected as the actuator for the SPD rocket’s payload deployment mechanism. Among the options considered, micro servos offer the best balance of positional control, response speed, mechanical simplicity, and electrical compatibility. Because payload release requires controlled rotation or latching rather than continuous or high-force linear motion, the servo’s built-in feedback system provides predictable and repeatable actuation without the need for external sensors or complex control circuitry. Micro servos also operate

efficiently from the rocket's 5-V rail and draw significantly less current than solenoids or linear actuators, reducing the load on the power system. Their compact size and lightweight construction make them suitable for integration into the constrained space of the payload bay. In contrast, solenoids and linear actuators would impose higher current demands, generate heat, and introduce mechanical bulk, while stepper and DC motors lack the inherent positional accuracy required for a secure latch or rotational release mechanism.

Power System Design

When deciding on the final power system design we balanced current capability, voltage stability, reliability, and ease of implementation within the constraints of a student-designed avionics platform. Early considerations included alkaline batteries, boost regulators, and hybrid buck-boost architectures; however, these approaches were found to be unsuitable due to limited current output, severe voltage sag under load, and high sensitivity to PCB layout. Feedback from previous senior design teams and faculty also emphasized the historically low success rate of TPS- and BQ-series boost regulators when used as primary power devices, especially when driving actuators or handling rapid load transients.

Specification	Linear Regulator	Switching Regulator		
		Buck	Boost	Buck-Boost
Efficiency	Low-Medium	High	High	High
Voltage	Step-Down	Step-Down	Step-Down	Step-Down
Heat	Medium-High	Low	Low	Low-Medium
Complexity	Low	Medium	Medium	High
Cost	Low	Medium	Medium	High

Table 03. Comparison of Regulator Design

To ensure robustness and predictable performance, the final power system adopts a 2-series lithium-ion or lithium-polymer battery pack as the main energy source. A 2S pack provides a nominal 7.4-volt supply, which is well above the required 5-volt and 3.3-volt rails and remains stable under high-current conditions. This higher input voltage allows the use of the LM2576-ADJ buck regulator, a proven and widely implemented device known for its tolerance to layout imperfections, thermal stability, and ability to supply several amperes of continuous current.

The final architecture uses two identical LM2576-ADJ regulators, one configured to output 5.0 volts for actuators, servos, and high-power peripherals, and the second configured to output 3.3 volts for the flight computer, sensors, GPS module, and telemetry electronics. Because both regulators share the same footprint and external component requirements—differing only in their feedback resistor values—the design is straightforward to implement and simplifies PCB layout, testing, and troubleshooting. This dual-buck configuration delivers clean, regulated power with sufficient headroom for transient loads during ascent and deployment, while avoiding the instability issues associated with boost-based solutions.

By selecting a 2S lithium power source and two LM2576-ADJ regulators, the final power system provides a reliable, redundant, and easily maintainable foundation for the rocket's avionics. This configuration ensures stable operation across all mission phases and offers a high margin of safety and electrical integrity for the project's critical subsystems.

Printed Circuit Board design

Designing reliable printed circuit boards (PCBs) is a central requirement of the PLEIADES avionics system because the electronics must survive launch loads, regulate power,

handle communication, and support all sensing and actuation functions without failure. The avionics stack is entirely dependent on stable PCB platforms that can withstand vibration, temperature changes, and rapidly fluctuating currents during flight. To meet these demands, the project employs modern EDA tools, robust connector strategies, and layout practices that follow widely accepted engineering standards for mixed-signal embedded aerospace systems.

KiCad was selected as the primary design tool because it provides all essential capabilities for PCB development, including schematic capture, component footprint management, routing, 3D visualization, and automatic generation of manufacturing files. KiCad's open-source nature allows for customized footprints, an important feature for modules such as the MS5611 altimeter, ICM-20948 IMU, MAX-M10S GPS receiver, and the RFM95W radio. The platform supports design rule checks that enforce track width, clearance, and copper thickness requirements necessary for a high-current power distribution board. Its integrated visualization tools enabled the team to verify that the PCBs fit within the avionics bay and aligned with mechanical mounting features, which is essential for flight readiness.

The connection strategy between PCBs must ensure mechanical stability and electrical reliability under vibration. Standard pin headers were used for most sensor-to-main-board connections because they are inexpensive, easy to assemble, and straightforward to rework. Their through-hole construction also provides greater mechanical strength. For high-current interfaces and connections exposed to higher mechanical loads—such as battery inputs, servo actuators, and the buzzer—the design uses JST-PH and JST-XH connectors. These connectors include positive locking mechanisms that prevent cables from disconnecting during sustained thrust or parachute deployment. RF communication requires its own class of connectors; U.FL pads and SMA connectors were chosen for LoRa and GPS antennas because they support

controlled-impedance routing and minimize insertion loss. By combining these connector types, each interface is optimized for its electrical and mechanical constraints.

Mechanical considerations heavily influence PCB construction. Because the avionics bay experiences strong axial loads and momentary lateral shocks, the PCBs use 1.6-mm FR-4 board thickness for rigidity and incorporate mounting holes that align with aluminum sled rails inside the rocket. Components that are relatively large or prone to vibration—such as inductors, electrolytic capacitors, and connectors—are reinforced with epoxy or hot glue. Cables are secured with heat-shrink tubing and tie-downs to eliminate strain at solder joints. In addition, conformal coating is applied to protect exposed copper from moisture during recovery. These techniques collectively ensure the avionics remain intact during both ascent and post-landing retrieval.

Electrical layout principles strongly influence PCB reliability. Because the avionics system includes both analog and digital elements, noise reduction and ground management are critical. Sensitive analog sensors such as the MS5611 altimeter and the IMU require clean grounds, so the sensor PCBs use isolated analog ground islands that return to the main ground plane at a single point. High-speed or noisy lines such as SPI to the LoRa transceiver, PWM outputs to actuators, and switching-regulator traces are routed away from these analog zones. Decoupling capacitors are placed as close as possible to sensor VCC pins, reducing voltage ripple and preventing transient-induced resets. These practices are essential for ensuring stable altitude readings and reliable inertial measurements.

The power system PCB requires careful consideration of copper geometry and return paths because actuator loads and radio transmissions can create sudden current spikes. A solid

copper ground plane reduces EMI and ensures low-impedance return paths. The LM2576 regulators used to generate the 5-volt and 3.3-volt rails require short, low-inductance loops around their switch nodes, so their layout is kept compact and isolated from analog areas. Wide copper pours are used for all high-current sections of the board, especially on the actuator supply rail. Bulk electrolytic capacitors near the regulator outputs absorb transient loads, while ceramic capacitors filter high-frequency ripple. By combining high-efficiency switching regulators with clean ground-plane organization, the power system maintains stability even during servo activation or telemetry bursts.

The RFM95W LoRa transceiver and MAX-M10S GPS module introduce RF-specific layout requirements. The LoRa transmitter operates at 915 MHz and requires a 50-ohm controlled-impedance trace between the transceiver and the antenna connector. To maintain this impedance, the RF trace is routed on a defined microstrip with a ground plane directly beneath it. Stitching vias placed along the RF path form a ground "fence" that reduces electromagnetic coupling to nearby traces. The GPS active antenna is positioned away from high-current regulators and switching inductors to prevent noise from saturating its internal low-noise amplifier. These layout considerations preserve RF performance and ensure that telemetry signals and GPS reception remain stable throughout flight.

Mechanical ruggedness and electrical stability are equally important in the design of the PCB interconnect architecture. The avionics system uses a hierarchical structure in which sensor PCBs plug into a central power and control board. This modular architecture simplifies debugging, replacement, and incremental testing. If a single board fails, the entire avionics system does not need to be rebuilt. It also reduces routing density on each board, minimizes cross-coupling between unrelated circuits, and reduces manufacturing risk.

The final selection of PCB technologies reflects the system’s mechanical, electrical, and RF requirements. KiCad provides a flexible and professional design environment capable of mixed-signal aerospace designs. Through-hole pin headers, JST power connectors, and U.FL/SMA RF connectors collectively support reliable, vibration-resistant interfacing. Dedicated sensor PCBs ensure clean measurement signals, while the main power PCB—built around dual LM2576 buck regulators—provides robust power rails with high noise immunity. The ground-station PCB benefits from native USB support available on the ESP32-S3 module and incorporates SMA connectors for strong antenna performance. Together, these PCB design strategies ensure that the avionics stack is electrically clean, mechanically robust, and fully capable of supporting the mission requirements of sequential payload deployment and telemetry transmission.

IV. System Requirements

The system requirements for the high-powered rocket were developed based on the user needs identified in the Needs Analysis and refined through benchmarking, advisor guidance, and mission constraints. These requirements outline the essential functions, performance thresholds, safety constraints, and physical limitations that the rocket must meet to successfully deliver its payloads and complete the mission. Requirements were categorized, assigned measurable target values, and paired with validation methods including inspection, testing, and analytical modeling. A complete listing of these requirements, including reference numbers and verification criteria, is presented in Appendix B - Table 16.

The rocket must meet several flight performance requirements to ensure a safe, stable, and successful mission. These include achieving a minimum ascent speed and altitude (e.g., 20

m/s minimum ascent speed and 1,320ft minimum altitude) and maintaining a stability margin within a predetermined range ($1.5\text{--}2.0 \pm 0.2$ calibers). Aerodynamic performance is constrained by a maximum drag coefficient, ensuring the rocket maintains sufficient velocity and altitude while minimizing energy losses. Proper attitude, orientation, and velocity tolerances are also required, particularly during payload deployment events, where limits such as 15° maximum attitude uncertainty and 20 m/s maximum speed uncertainty help ensure controlled operation.

The propulsion system must generate adequate impulse to conduct the launch. Motors in the range of 320–640 Ns satisfy the mission requirement while aligning with available launch infrastructure. The rocket must also be compatible with the designated launch system, including the use of standard launch rail guides.

To support mission-critical avionics, the electrical system requirements specify that the onboard battery must provide at least 7.5 Wh of usable energy over the full flight duration. Power regulation must ensure stable operation of sensors, microcontrollers, actuators, and telemetry equipment. The avionics bay must house modular electronics securely while supporting easy access for pre-flight assembly and diagnostics.

The sensor and telemetry subsystem must acquire, transmit, and log accurate real-time flight data. Requirements include achieving sampling rates of at least $100\text{ Hz} \pm 10\text{ Hz}$ and maintaining measurement uncertainties within acceptable bounds such as:

- Location uncertainty: $\leq 10\text{ m}$
- Altitude uncertainty: $\leq 10\text{ m}$
- Time uncertainty: $\leq 1\text{ s}$
- Attitude uncertainty: $\leq 15^\circ$

- Speed uncertainty: ≤ 20 m/s

Telemetry data must be transmitted with sufficient speed to support real-time monitoring; onboard storage (≥ 10 GB) ensures that data is preserved for post-flight analysis in the event of transmission loss. Antennas and communication modules must be selected and configured to meet these real-time data transfer requirements.

To ensure safe and reliable payload deployment, the rocket must carry three payloads and release them at predetermined altitudes. Deployment must occur within specified timing constraints, including a maximum deployment delay of 10 seconds, a minimum spacing of 3 seconds between successive payloads, and minimal attitude shock ($\leq 15^\circ$) during release. The deployment system must also maintain a factor of safety between $2-4 \pm 0.5$ for both the rocket structure and the release mechanism.

Descent and recovery requirements include limiting descent speed to ≤ 11 m/s to ensure safe landing and preventing the rocket from contacting the ground horizontally outside the acceptable orientation limits (e.g., (0,0,0) to (180,10,10)). Dynamic pressure tolerance ($6 \text{ kPa} \pm 1 \text{ kPa}$), structural integrity, and compatibility with standard recovery systems are essential for ensuring the vehicle remains intact for post-flight data extraction and reuse.

Operational and environmental constraints include limiting launch operations to suitable weather conditions—for example, the rocket must not be launched into a cloud—to ensure safety and compliance with launch regulations. Mechanically, the rocket must be easily assembled within 30 minutes and constructed modularly with at least two primary sections to allow access to internal components.

Finally, several manufacturing and cost constraints guide material and subsystem choices. All components must be producible using available machines and fabrication tools, ensuring manufacturability within the university environment. Total system cost must remain below the target budget of \$700, supporting project affordability and educational constraints.

Together, these system requirements translate user needs into measurable engineering specifications. They provide a structured foundation for subsystem design, integration, testing, and verification. The full list of detailed requirements, tolerances, and validation methods is presented in Appendix B - Table 16.

V. Concept Generation and Selection

With the engineering requirements defined, as well as customer needs and wants, brainstorming began on what preliminary form the project would take. The team began by making a web diagram encompassing each element of the rocket's flight with multiple solutions to each problem. From there, Pugh matrices were crafted to compare these solutions and choose the best one.

The first step in the concept generation process was to determine how the team wanted to think of and expand upon different ideas. Beginning with the main concept — The Rocket — the team developed a web of the most important possible variations of creating a rocket. These were mostly focused on the overall design, as this would be the first necessary decision to make. Figure 1 below shows the full brainstorming web diagram the team laid out.



Figure 02. Single Stage Focus Web Diagram Brainstorm

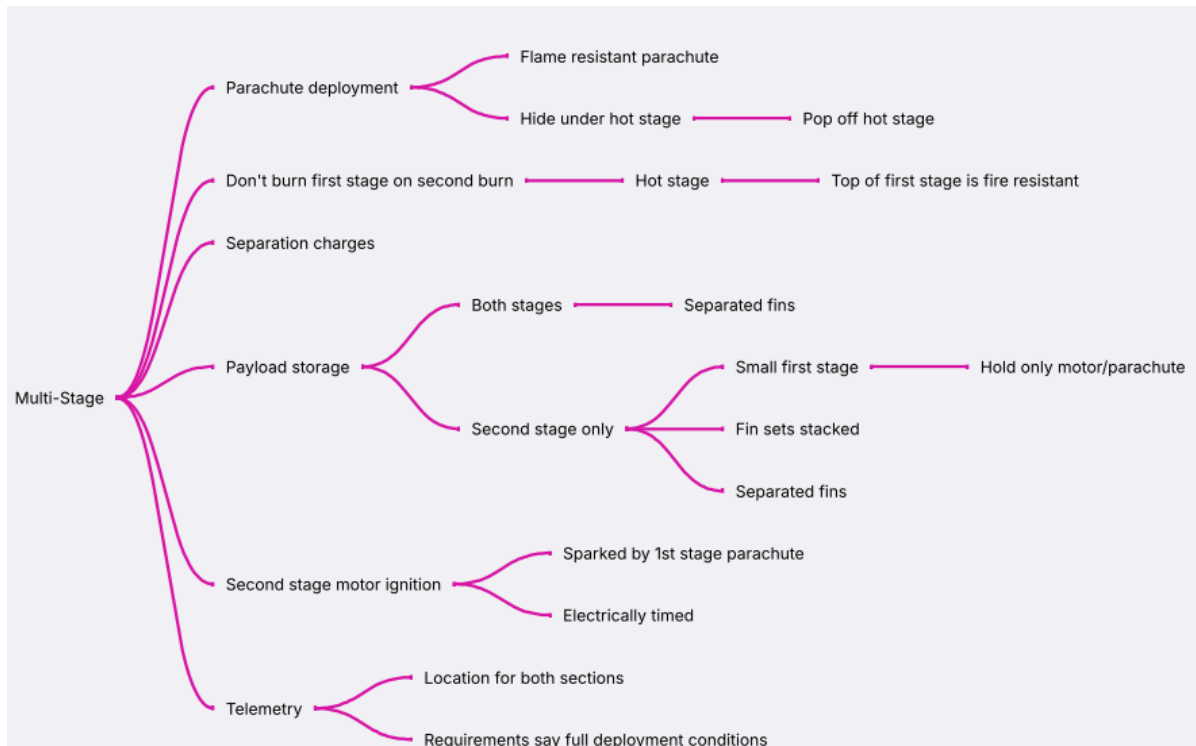


Figure 03. *Multi-Stage Focus Web Diagram Brainstorm*

Within these two categories, the impact of motor configuration on payload and parachute deployments was analyzed in order to get a good overview of what options the team would have. One option was sequential deployment of payloads, either through vertical deployment via separations or a hatch bay that would launch payloads out the side of the rocket. Multiple configurations of each were considered.

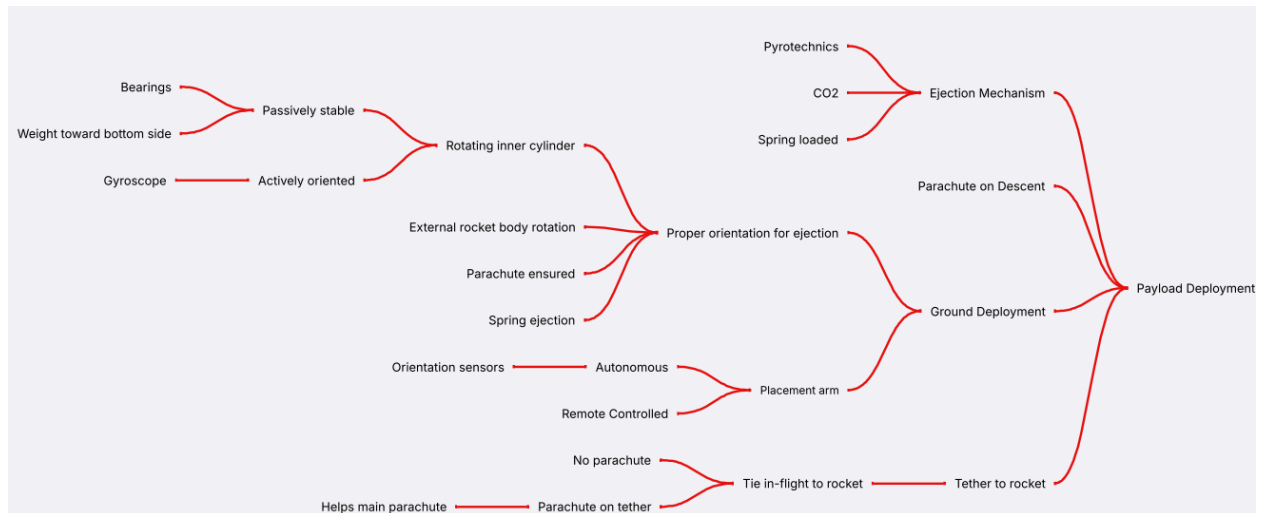


Figure 04. *Payload Deployment Focus Web Network Brainstorm*

A few other smaller concepts were also expanded upon, such as the fin design and nose cone, but these had very few variations and were easily fully generated without much to choose from.



Figure 05. *Nose Cone Focus Web Network Brainstorm*

The final major system was the electrical system. The team first started by creating a list of every electronic system before thinking of which sensors or motors would fulfill each duty best. Then, any crossovers were eliminated to leave a completed list. Multiple ideas were able to

be combined in several sections, yielding a large but not excessive list of components required by the rocket.



Figure 06. *Electrical Focus Web Diagram Brainstorm*

Taking the ideas from these brainstorming diagrams, the team compiled Pugh matrices to narrow the scope of the project to a preliminary solution. For each Pugh matrix, the procedure

followed as shown below. Key criteria were identified to rate the motors on. These key criteria were given an importance score out of 10 to represent their relative weight in decision making. Finally, a benchmark option was chosen to which the other choices would be compared to. If a design choice outperforms the benchmark in a key criteria area, it is marked with a “+.” If a design choice has no significant difference from the benchmark, it is marked with “S.” And if a design choice is worse than the benchmark option, it is marked with a “-.” These +, -, and S’s are tallied and weighted according to the importance rating of each criteria. The option with the greatest total was selected as the best option for the preliminary design stage (if all options are negative, the benchmark is the best).

Propulsion Pugh Matrix				
Key Criteria	Importance Rating	Solution Alternatives		
		Single Motor	Motor Cluster	Two Stage Motor
Concept Selection Legend Better + Same S Worse -				
Cost	7		-	-
Weight	2		-	-
Ease of Assmby	3		-	-
Complexity of Ignition	8		-	-
Reliability	5		S	-
Safety to Rocket	2		-	-
Safety to Surroundings	10		S	+
Portability	2		S	S
Sum of Positives			0	1
Sum of Negatives			5	6
Sum of Sames			3	1
Weighted Sum of Positives			0	10
Weighted Sum of Negatives			22	27
TOTALS			-22	-17

Table 04. Propulsion Configuration Pugh Matrix

By applying the Pugh matrix to the motor configuration options, a single stage, single motor configuration is plainly the best option. With the weighted rankings, it ended 17 points clear of the next best option. For the rest of the Pugh matrices, see Appendix B.

After selecting a single-stage, single motor rocket, the team looked at mechanisms for the sequential deployment of payloads. The primary two options that were generated were to use either vertical deployment via separations or a hatch bay that would launch payloads out the side of the rocket. In the case of a vertical deployment, the rocket would separate and deploy a parachute first. Then while under the chute, it would release a payload from inside the body tube. In the case of a hatch bay, the payloads could be ejected at any time, though still ideally under chute. A charge would force some sort of hatch or panel cover off, and the payload would be deployed from the side of the body tube. After analysis, a vertical deployment was selected.

The final physical system is the ground ejection. This payload will simulate a rover, and as such must be deployed only after landing. The main concept here was how to ensure the rocket was properly oriented after landing for the payload deployment to work, and a passive counter-weight system that would keep the mechanism upright was selected. Once the rocket had landed and the counter-weight had settled, the payload could be deployed by a spring.

Altitude, speed, and attitude would need to be measured, and the team decided that combining all three with an Inertial Measurement Unit (IMU) was the best plan. Location can be determined with both a beeper for audio locating or a GPS for more precise measurements. The final plan was to use both a GPS and beeper for the rocket, and individual beeper sensors for each payload. Looking at telemetry, a transmission method was required to get data from the rocket to the controller. This will be done with an antenna, and through analysis, an active

antenna was chosen. More complex systems such as power production and supply were investigated further to get a list of all necessary components, and the team selected an Alkaline battery as a power source.

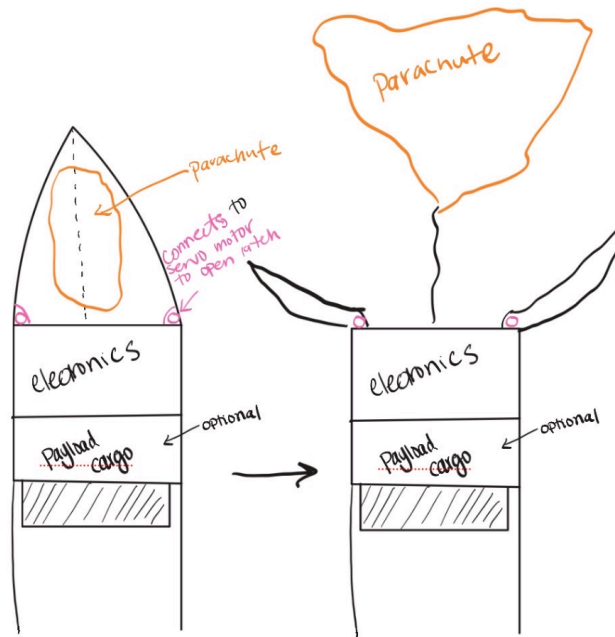


Figure 07. *Nose cone Design 1*

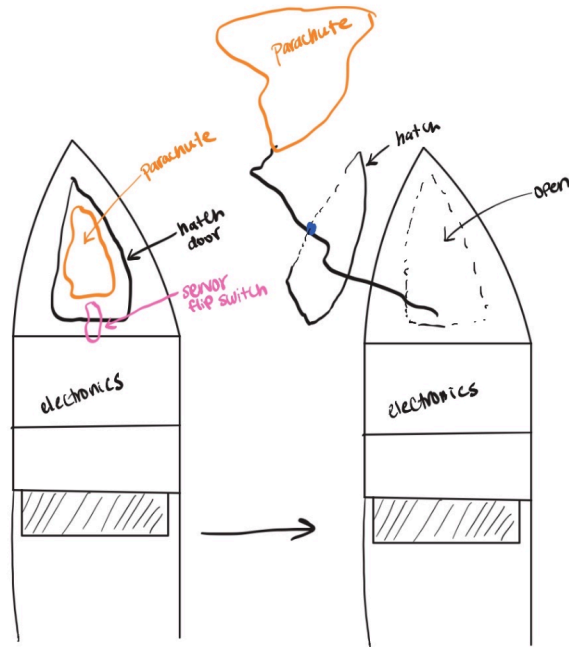


Figure 08. *Nose cone Design 2*

To achieve the goal of releasing payloads vertically under drogue, the nose cone of the rocket will first need to be ejected. Two potential mechanisms for ejecting the nose cone are shown in the figures above. The first concept involves splitting the nose cone in two pieces that can be assembled and connected by servo motors at the base. Once apogee is reached, the electronics will send a signal to trigger the servo motors to separate the two nose cone halves and deploy the nose cone parachute. The two separated halves of the nose cone will still remain connected to its base for recoverability.

The second concept differs from the first in that only one servo motor is mounted on the exterior of a hatch-style bay door. Similar to the first release mechanism, the servo motor will initiate once apogee is reached, releasing the hatch and allowing the door to fall off the nose cone entirely. The weight and motion of the hatch would then pull the parachute out of the nose cone for deployment. The hatch would be attached to the shock cord to ensure it is not lost during descent.

Additionally, the team considered placing another small payload within the nose cone, which is why there is a potential dedicated space for the payload cargo underneath the electronics.

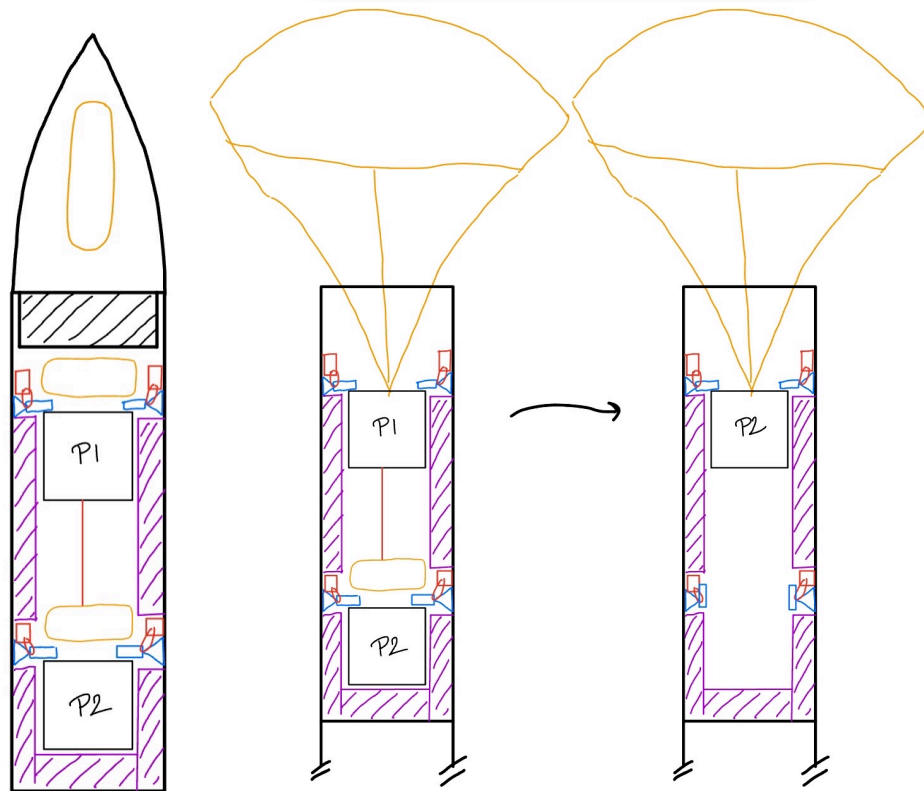


Figure 09. *Payload Mechanism Design*

The exact mechanism for releasing the payloads from the payload bay under drogue is not yet decided though one of the ideas that the team devised is illustrated in the figure above. In this design, the two payloads are connected and aligned along a linear rail guide (depicted in purple). The blue levers interface with an electrically driven release mechanism (depicted in red) to hold the payloads in the body tube while the parachute is inflated. When the deployment sequence is initiated, the electrical mechanism actuates the blue levers, causing them to retract

and sit flush against the linear rail. This motion prevents the latch from potentially getting stuck on the payload or parachute during deployment.

The goal of this design is to deploy the first payload (P1) soon after the nose cone deployment. The bottom section of P1 briefly grips the second payload's parachute, thereby assisting in the deployment of the second parachute (P2) and leaving the rocket on a separate drogue after P1 is deployed. P2 would then release in a similar manner.

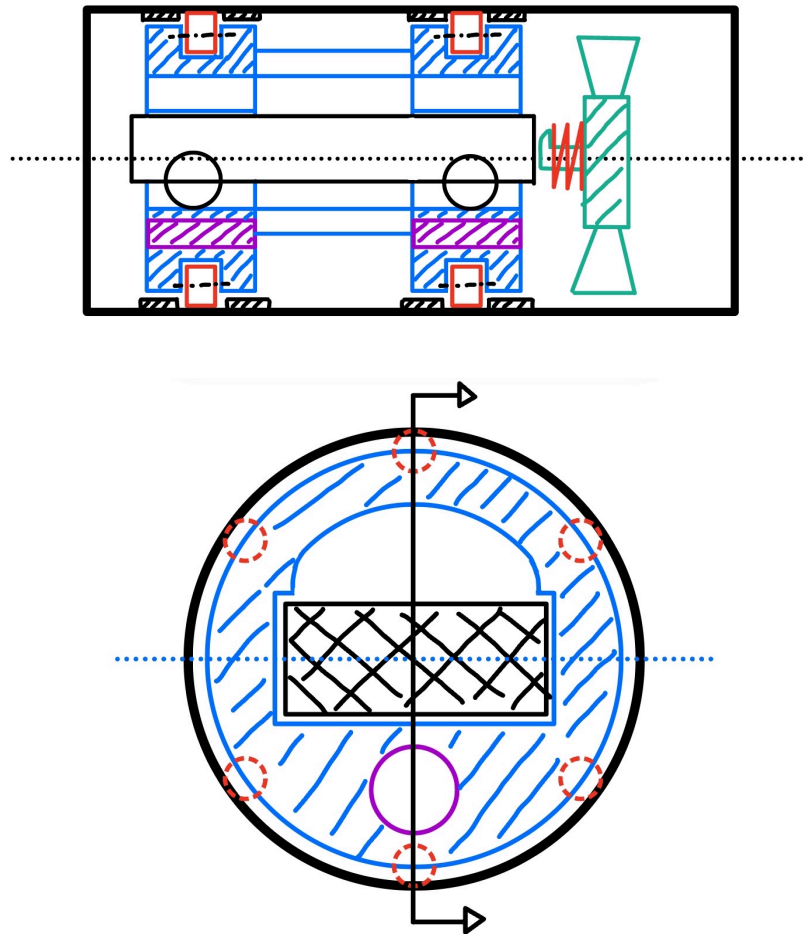


Figure 10. Ground Deployment Mechanism

The basic theory of the ground deployment system is to have a cylindrical assembly (shown in blue) resting on bearing wheels (shown in red) with a counter weight at the bottom (shown in purple) to ensure the orientation of the ground payload always remains vertical. The cylindrical assembly would rotate freely within the body tube, thus eliminating concerns over the rocket landing in a particular orientation. The left end in Figure 10 would be open once the payload is on the ground to allow the spring mechanism in green to push the payload out of the body tube. The major concerns with this type of deployment involve ensuring the payload is deployed softly, not thrown out at random. Testing will need to be performed as different solutions to this issue arise, but some potential fixes are a ramp where the payload ejects, using a lighter stiffness spring to avoid excessive shock, and exchanging the spring mechanism for one that can place the payload on the ground.

VI. Preliminary Engineering Analysis

For this project, the conducted analysis focused on overall performance given a motor class limit and potential total weights. In other words, due to the group's restriction to I-class motors (or lower) and the deployment system the team plans to incorporate, the question of "power/weight ratio" is key. A minimum altitude must be set to ensure the proper deployment of all payloads. If the rocket's weight is too great, the team risks not having enough room to deploy all the intended payloads before landing. There is no concern for maximum allowable altitude, so the focus was on achieving the minimum.

To get a rough estimate of how motor constraints could affect the altitude, the group first needed to gather some fundamental information, such as ideal performance data. Raw calculations were done first, and the list of assumptions for this was as follows:

- I-class motor with the most total impulse it can have within its classification (640 N s)
- Negligible friction and drag effects (unrealistic, but gives an idea of the maximum height for an impulse).
- A relatively long burn time (8.5s).
- Completely vertical launch trajectory (impossible, but fine for an estimate).
- Constant mass (which works against performance).
- Reliable burn time (solidly 8.5s, no longer or shorter).
- Constant thrust curve with negligible variation.

The velocity at burnout achieved after motor thrust is complete can be determined starting with the linear momentum and impulse equations. Because the rocket is stationary before launch, its initial momentum is 0.

$$L_{tot} = mv$$

$$I = \Delta L = L_f - L_i$$

$$I = L_f - 0 = L_f = L$$

$$I_{tot} = L_{tot} = mv \quad (1)$$

$$L_{tot} = L_{motor} + L_{grav}$$

$$L_{motor} = I_{motor}$$

$$L_{grav} = I_{grav} = \int_0^t F_{grav} dt = - \int_0^t mg dt = - mgt$$

$$L_{tot} = mv_b = I_{motor} - mgt \quad (2)$$

$$v_b = \frac{I_{motor}}{m} - gt \quad (3)$$

Where t is the burn time, I_{motor} is the motor impulse, and v_b is the velocity at burnout.

Because the thrust force of the rocket and the rocket mass are assumed to be constant, the height the rocket achieves across the motor's thrust can be determined as follows.

$$F = \frac{dL}{dt}$$

$$F = \frac{d}{dt} (mv)$$

$$F = ma$$

$$a = \frac{d^2x}{dt^2}$$

$$x = \int_0^t \int_0^t a \, dt^2 = \int_0^t \int_0^t \frac{F}{m} \, dt^2 = \frac{F}{m} * \int_0^t \int_0^t dt^2$$

$$x = \frac{F}{m} * \frac{1}{2}t^2$$

Position (x) in this equation can be replaced by height (h_b) as the analysis is done in the y-direction.

$$h_b = \frac{F}{m} * \frac{1}{2}t^2 \quad (4)$$

$$I_{motor} = \int_0^t F_{motor} \, dt = F_{motor} \int_0^t dt = F_{motor} t$$

$$F = F_{motor} + F_{grav}$$

$$F = \frac{I_{motor}}{t} - mg$$

$$h_b = \frac{1}{2}t^2 * \left(\frac{I_{motor}}{mt} - g \right) \quad (5)$$

m (kg)	I_{motor} (N*s)	I_{grav} (N*s)	L_{tot} (N*s)	h_b (m)	v_b (m/s)
1.5	640	125.078	514.923	1458.947	343.282
2	640	166.770	473.230	1005.614	236.615

2.5	640	208.463	431.538	733.614	172.615
3	640	250.155	389.845	552.280	129.948
3.5	640	291.848	348.153	422.757	99.472
4	640	333.540	306.460	325.614	76.615
4.5	640	375.233	264.768	250.058	58.837
5	640	416.925	223.075	189.614	44.615
5.5	640	458.618	181.383	140.159	32.979
6	640	500.310	139.690	98.947	23.282
6.5	640	542.003	97.998	64.075	15.077
7	640	583.695	56.305	34.185	8.044
7.5	640	625.388	14.613	8.280	1.948

Table 05. *Burnout Calculations without Drag*

The left column for mass was generated arbitrarily, demonstrating how the rocket's performance varies over a range of values. The starting point was 1.5 kg, as that is the minimum mass requirement for attaining an L1 Certification, and then it was incremented by 0.5 kg per row. The velocity after burnout under these conditions was tabulated because apogee altitude is not entirely dependent on burnout height, but also on the additional height the rocket acquires as its momentum carries it further into the sky. The constant-acceleration kinematic relation was used to find the additional altitude attained after burnout:

$$v_f^2 = v_b^2 + 2a\Delta h$$

v_f^2 is velocity at apogee, v_b^2 is velocity at burnout, a is acceleration, and Δh is change in altitude.

This expression was rearranged with the following conditions/terminology:

- Velocity at apogee is 0.
- The only acceleration acting on the rocket is gravity (-g).

Yielding,

$$0 = v_b^2 - 2g\Delta h$$

$$\Delta h = \frac{v_b^2}{2g} \quad (6)$$

$$h_{tot} = h_b + \Delta h \quad (7)$$

The total height achieved for each mass is tabulated in the table below.

m (kg)	h_b (m)	Δh (m)	h_{tot} (m)	h_{tot} (ft)
1.5	1458.947	6006.234	7465.181	24493.258
2	1005.614	2853.550	3859.164	12661.917
2.5	733.614	1518.651	2252.265	7389.682
3	552.280	860.681	1412.962	4635.928
3.5	422.757	504.317	927.074	3041.730
4	325.614	299.177	624.791	2049.939
4.5	250.058	176.443	426.502	1399.352
5	189.614	101.453	291.066	954.988
5.5	140.159	55.433	195.592	641.737
6	98.947	27.627	126.574	415.289
6.5	64.075	11.585	75.661	248.242
7	34.185	3.298	37.483	122.981
7.5	8.280	0.193	8.474	27.803

Table 06. *Height Calculations without Drag*

The team was curious about the estimated maximum mass the vehicle could possess to reach a total altitude of 5280 feet (one mile). To find that value, an interpolation between 2.5 kg and 3 kg was performed, yielding a maximum mass of 2.883 kg in a drag-free, ideal environment.

This mass estimation told the team that, given the nature of the project revolving around advanced deployment systems (adding substantial weight to the vehicle), the altitude

expectations may have had to be adjusted. To confirm this pivot, another iteration of calculations was conducted, this time incorporating a drag element.

The drag figures were generated assuming typical L1 simulations, yielding a drag impulse of 5% of the total impulse $I_D = 0.05 * I_{tot}$. A drag force was derived by dividing the

drag impulse by the burn time $D = \frac{I_D}{t}$. This approach assumed a constant, average drag force.

An average acceleration can be determined by a sum of forces.

$$\Sigma F = ma$$

$$F_{motor} - mg - D = ma$$

$$\frac{I_{motor}}{t} - mg - D = ma$$

$$a = \frac{I_{motor}}{mt} - g - \frac{D}{m} \quad (8)$$

Where a is the average acceleration during the burn, I_{motor} is the motor impulse, t is the burn time, D is the average drag during the burn, and m is the mass. The drag force, average acceleration, and resulting burnout velocity are tabulated below.

m (kg)	D (N)	a (m/s ²)	v_b (m/s)
1.5	30.290	38.367	326.118
2	27.837	26.445	224.784
2.5	25.385	19.292	163.984
3	2.293	14.524	123.451
3.5	20.480	11.117	94.499
4	18.027	8.563	72.784
4.5	15.575	6.576	55.895
5	13.122	4.986	42.384
5.5	10.670	3.686	31.330
6	8.217	2.602	22.118

6.5	5.765	1.685	14.323
7	3.312	0.899	7.641
7.5	0.860	0.218	1.851

Table 07. Burnout Calculations with Drag

This new velocity, accounting for the drag force, demonstrates how the drag force has a greater impact on rockets with less mass, becoming less significant with heavier weights. The resulting burnout height can be calculated in the same way as the drag-free approximation. To determine the apogee height, the glide height Δh must be found using another sum of forces.

$$\begin{aligned}
 \Sigma F_{glide} &= ma_{glide} \\
 -mg - D &= ma_{glide} \\
 D &= 0.05 * \frac{I_{tot}}{t_{glide}} \\
 I_{tot} &= I_{grav} = mgt_{glide} \\
 D &= 0.05 * \frac{mgt_{glide}}{t_{glide}} \\
 D &= 0.05 * mg \\
 -mg - 0.05 * mg &= ma_{glide} \\
 a_{glide} &= -1.05g \quad (9)
 \end{aligned}$$

Thus, the average deceleration of the rocket is 1.05 times the force of gravity when the drag force is included with the L1 simulation assumption. This average deceleration yields the following height calculations.

m (kg)	h_b (m)	Δh (m)	h_{tot} (m)	h_{tot} (ft)
1.5	1386.000	5162.501	6548.500	21485.630
2	955.333	2452.694	3408.028	11181.738

2.5	696.933	1305.317	2002.250	6569.382
3	524.666	739.776	1264.443	4148.636
3.5	401.619	433.473	835.092	2739.935
4	309.333	257.150	566.483	1858.631
4.5	237.555	151.657	389.213	1277.006
5	180.133	87.201	267.334	877.123
5.5	133.151	47.646	180.797	593.195
6	94.000	23.746	117.746	386.323
6.5	60.872	9.958	70.829	232.391
7	32.476	2.834	35.310	115.853
7.5	7.866	0.166	8.033	26.355

Table 08. *Height Calculations with Drag*

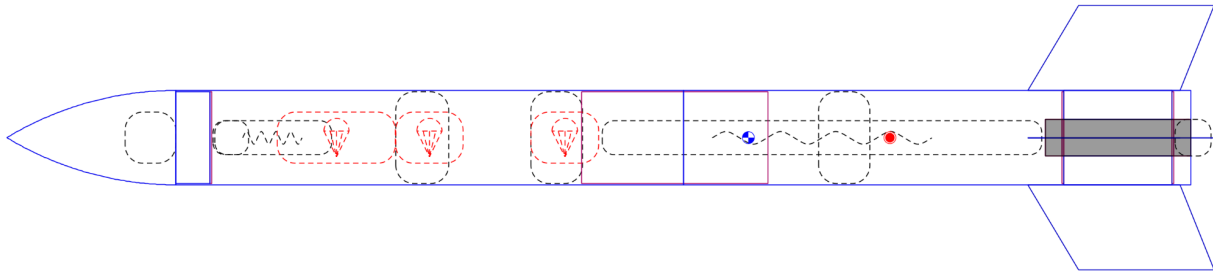
By interpolation, the rocket must have a maximum mass of 2.766 kg to attain an altitude of 5280 feet, once again a worryingly low number. For a quarter mile, the maximum weight is 4.463 kg, which is far more achievable.

The assumptions listed earlier played a huge role in the accuracy of the calculation results. To determine how big a role they played, the hand calculations were compared to several simulations in OpenRocket involving a variety of motor configurations. The simulations were run at 8.58 lbs, or 3.892 kg.

Motor Type	Velocity off rod	Apogee	Max. velocity	Max. acceleration	Time to apogee	Flight time
[I65-10]	29.7 ft/s	1960 ft	264 ft/s	133 ft/s ²	12.8 s	79.1 s
[I500-14]	53.2 ft/s	1911 ft	471 ft/s	471 ft/s ²	10.1 s	83 s
[I284-14]	49.4 ft/s	1862 ft	418 ft/s	377 ft/s ²	10.2 s	81.8 s
[I132-14]	30.3 ft/s	1843 ft	324 ft/s	150 ft/s ²	11.4 s	68.8 s
[I280-14]	41.2 ft/s	1691 ft	393 ft/s	256 ft/s ²	10 s	74.3 s
[I180-10]	46.2 ft/s	893 ft	230 ft/s	305 ft/s ²	7.64 s	41.1 s

Table 09. Open Rocket Initial Simulations

SD1_Rev1	Stability: 1.51 cal / 11.7 %
Length 71.375 in, max. diameter 5.54 in	● CG: 43.845 in
Mass with no motors 6.93 lb	● CP: 52.215 in
Mass with motors 8.58 lb	at M=0.300



Apogee: 1964 ft
Max. velocity: 264 ft/s (Mach 0.238)
Max. acceleration: 133 ft/s²

Figure 11. Preliminary Rocket Calculations

Although the specific arrangement and assembly of the rocket were not fully defined, estimates were made based on the potential motors the team will be using for the vehicle. The simulations confirmed the suspicion of apogee expectations, aligning with the general conclusion drawn from the raw calculations (2049ft by hand calculations). However, although some simulated apogee values were substantially lower than initial estimates, the simulated altitude is high enough that there will be no concern during payload deployment.

By employing fundamental dynamics principles and making assumptions where the team could, a surface-level analysis helped validate altitude expectations and general kinematic

behavior of the launch vehicle. Moving forward, this data will serve as the foundation for further evaluations regarding motor performance, propulsion system design, and hardware assembly.

VII. *Gantt Chart and Major Deliverables*

The team developed an extensive Gantt chart derived from a comprehensive block-diagram flowchart built during the planning process. The flowchart subdivided each subsystem—mechanical, electrical, software, testing, and documentation—into actionable tasks with explicit dependencies. The Gantt chart encompasses the whole project timeframe and offers a daily delineation of tasks for all team members.

Gantt Chart	9/12	9/13	9/14	9/15	9/16	9/17	9/18	9/19	9/20	9/21	9/22	9/23	9/24	9/25
Task:	F	S	S	M	T	W	R	F	S	S	M	T	W	R
M-1 Requirement														
Team Contract														
TSM D&C														
Component Breakdown														
Function Breakdown														
Come up with Team Name														

Gantt Chart	10/2	10/3	10/4	10/5	10/6	10/7	10/8	10/9	10/10
Task:	R	F	S	S	M	T	W	R	F
M-2 Requirement									
House of Quality (HoQ)									
1. Customer Requirements									
2. Importance Weights									
3. Engineering Characteristics									
4. Correlation Ratings									
5. Benchmark Satisfaction Ratings									
6. Benchmark Performance									
7. New Product Performance Targets									
8. Coupling									

Gantt Chart	10/10	10/11	10/12	10/13	10/14	10/15	10/16	10/17	10/18	10/19
Task:	F	S	S	M	T	W	R	F	S	S
Write Engineering Requirements										
Write M2 Report										
Section 1										
Section 2.1										
Section 2.2										
Section 2.3										
Section 2.4										
Section 2.5 (A)										
Section 2.5 (B)										

Gantt Chart	10/30	10/31	11/1	11/2	11/3	11/4	11/5	11/6	11/7	11/8	11/9
Task:	R	F	S	S	M	T	W	R	F	S	S
M-3 Requirement											
Gantt Chart Organization											
Major Deliverables											
Concept Generation											
Concept Selection											
Preliminary Engineering Design											
System Concept Evaluation Plans											
FMEA Analysis											

Gantt Chart	11/8	11/9	11/10	11/11	11/12	11/13	11/14	11/15	11/16	11/17	11/18	11/19	11/20	11/21	11/22	11/23	11/24	11/25
Task:	S	S	M	T	W	R	F	S	S	M	T	W	R	F	S	S	M	T
M-3 Requirement																		
Major Deliverables																		
Preliminary Engineering Design																		
System Concept Evaluation Plans																		
FMEA Analysis																		
ABET Criteria																		
Significant Accomplishments																		

Gantt Chart	11/10	11/11	11/12	11/13	11/14	11/15	11/16	11/17	11/18	11/19	11/20	11/21	11/22	11/23	11/24	11/25
Task:	M	T	W	R	F	S	S	M	T	W	R	F	S	S	M	T
Write M3 Report																
Initial Information																

Needs & Requirements																			
Tech Assessment (shared)																			
Concept Generation																			
Preliminary Engineering Analysis																			
Gantt and Evaluation																			
FMEA Report																			
Work, Conclusion, & References																			

Gantt Chart	12/5	12/6	12/7	12/8	12/9	12/10	12/11	12/12	12/13	12/14	12/15	12/16	12/17	12/18	12/19	12/20	12/21	12/22	12/23	12/24
Task:	F	S	S	M	T	W	R	F	S	S	M	T	W	R	F	S	S	M	T	W
M-4 Requirement																				
Component Research																				
CAD Model of Components																				
Cost Analysis																				
Electrical Model																				
ANSYS Model																				
CAD Assembly and Integration																				
Open Rocket System Analysis																				
Coding and Web Development																				

Gantt Chart	12/25	12/26	12/27	12/28	12/29	12/30	12/31	1/1	1/2	1/3	1/4	1/5	1/6	1/7	1/8	1/9	1/10	1/11	1/12	1/13	1/14	1/15	1/16	1/17	1/18
Task:	R	F	S	S	M	T	W	R	F	S	S	M	T	W	R	F	S	S	M	T	W	R	F	S	S
M-4 Requirement																									
Component Research																									
CAD Model of Components																									
Cost Analysis																									
Electrical Model																									
ANSYS Model																									
CAD Assembly and Integration																									

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Table 10. Sectioned Gantt Chart

Major Deliverables	Due Date	Brief Description
M3 Report	11/25/2025	
Long Term Project Plan	12/15/2025	Plan out project lifespan through SD2
M4 Report	1/18/2026	
Working CAD Model	1/12/2026	CAD model sufficient to begin testing
Electrical Model	1/12/2026	Electrical schematic sufficient to begin testing
Detailed FMEA	2/4/2026	Full revised FMEA with new testing knowledge

M5 Report	2/8/2026	
1st Launch Attempt	3/14/2026	First launch window for rocket
M6 Report	4/27/2026	

Table 11. Major Deliverables

VIII. System Concept and Evaluation Plans

To evaluate the feasibility and reliability of the chosen concept, the team conducted a systematic risk assessment, concentrating on potential failure spots that may affect execution. Each risk was assessed according to probability, severity, and detectability.

Risk	Potential Impact	Likelihood	Mitigation Strategy
Component failure/availability delays	Project timeline setbacks	Medium	Order backup components; verify vendor lead times early
Integration issues between subsystems	System instability/malfunction	Medium–High	Define interfaces early, perform subsystem tests before integration
Software bugs or timing errors	Incorrect system behavior	High	Incremental testing; code reviews; simulation where possible
Mechanical tolerance issues	Physical misalignment subsystem failure	Medium	CAD verification, early prototyping, and tolerance checks
Power limitations and inefficiencies	System resets, overheating	Low–Medium	Include power budget analysis and safety margin
Environmental/field conditions	Reduced performance or failure	Low–Medium	Test in representative conditions; stress testing

Table 12. Risk Assessment

The team will apply the following structured strategy to assess and test the chosen concept, taking into account the identified concerns:

1. Testing at the Subsystem Level

Before integration, each subsystem (mechanical structure, microcontroller code, sensors, communication modules, etc.) will undergo independent testing to identify errors and confirm fundamental functionality.

2. Validation of Interfaces

To avoid integration conflicts, early testing will be done on subsystem interfaces, including signal levels, communication protocols, mounting locations, and power needs.

3. Modeling and Simulation

Simulations will be utilized where appropriate to identify design faults early on: Simulation of electricity (LTspice, Multisim), Simulation of microcontroller firmware logic, and Checks for mechanical stress and tolerance in CAD

4. Integration in Steps

Subsystems will be integrated gradually rather than all at once, with each integration phase being followed by performance and stability checks.

5. Prototyping for Risk Reduction

The assessment's highest-risk areas—such as mechanical actuation strength, sensor precision, and communication reliability—will be the focus of early prototypes.

6. Validation of the entire system

After integration, the entire system will go through: Functional testing from start to finish, Testing for durability and stress, Validation of performance under anticipated use scenarios.

7. Review and Documentation

Throughout the project, the team will update risk levels, record test findings, and make necessary design adjustments.

IX. Failure Modes and Effects Analysis

Item / Function	Potential Failure Mode(s)	Potential Effect(s) of Failure	Sev	Potential Cause(s)/Mechanism(s) of Failure	Prob	Current Design Controls	Det	RPN	Recommended Action(s)
Propellant Grains	Failed/improper ignition, incomplete combustion	Rocket does not reach intended altitude	7	Poor mixing from manufacturer or Improper care of grains (i.e. proper storage, no moisture)	2	None	10	140	Focus on storing grains properly (according to manufacturer specifications)
Motor casing	Cracks in wall	Damage to rocket and surroundings	10	Improper material selection or excessive stress or manufacturing defect	1	Visual Inspection	2	20	Regularly check integrity and replace if needed
Motor casing	Bolt shear	Damage to rocket and surroundings	10	Improper material selection or excessive stress/fatigue or manufacturing defect or stripped threads	1	None	10	100	Run detailed simulations
Motor casing	Bolt tearout	Damage to rocket and surroundings	10	Improper material selection or excessive stress/fatigue or manufacturing defect	1	None	10	100	Run detailed simulations
Motor casing	Thermal deformation	Reduction in structural integrity, potential damage to rocket	9	Improper material selection or excessive stress or manufacturing defect	1	None	10	90	Ensure phenolic liner is intact
Nozzle Cap	Cracks in cap	Grains fall out or motor does not produce the intended thrust	3	Improper material selection or excessive stress or manufacturing defect	3	Visual Inspection	1	9	Be cautious while handling motor
Nozzle Cap	Cap falls off during storage	Grains fall out or motor does not produce the intended thrust	3	Improper fit or manufacturing defect	2	Visual Inspection	1	6	Ensure cap is properly attached
Aft Closure	Tearout	Decrease in thrust	7	Improper material selection or excessive stress/fatigue or manufacturing defect or stripped threads	1	None	10	70	Run detailed simulations
Aft Closure	Seal is not sufficient	Shifting of lining within casing	4	Improper seating	2	Visual Inspection	1	8	Ensure aft closure is properly attached
Aft Closure	Cracks/Damage	propellant/thrust leakage	4	Improper material selection or excessive stress or manufacturing defect	2	Visual Inspection	2	16	Check integrity and replace if needed
Nozzle	Broken nozzle	Loss of thrust and potential misfire	4	Improper material selection or excessive stress or manufacturing defect	2	Visual Inspection	1	8	Check integrity and replace if needed
Motor O-rings	O-ring snaps	Improper sealing (leading to leaks, higher temps, etc)	9	Environment/weather	3	Visual Inspection	4	108	Check integrity and replace if needed
Motor O-rings	Cracking	Improper sealing	9	Improper material selection or excessive stress or manufacturing defect or too old/stiff	1	Visual Inspection	2	18	Check integrity and replace if needed
Aft Insulator	Damaged insulation	Damage to aft o-ring and other adjacent components	5	Manufacturing defect	1	None	10	50	None
Phenolic Liner	Holes in the lining	Leaks of propellant grain, decreasing thrust	9	Manufacturing defect	1	None	10	90	Proper storage to avoid damage
Forward Seal Disc	Damaged disk	Damage to the upper sections of the motor, possible combustion of the ejection charge	5	Manufacturing defect	1	None	10	50	None
Forward Seal Disc	Seal is not sufficient	Damage to the upper sections of the motor, possible combustion of the ejection charge	4	Improper seating	1	None	10	40	None
Aft Delay Spacer	Incorrect Size	Delay charge could move around, incorrect ignition	3	Assembly defect	1	Visual Inspection	1	3	If team changes delay, ensure spacer reflects the change
Delay insulator	Damaged insulation	Damage to delay element and other adjacent components, possible early ignition of the ejection charge	2	Manufacturing defect	1	None	10	20	None
Delay element	Failed/improper ignition	Parachute deploys at an incorrect time or not at all	10	Poor mixing from manufacturer or delay spacer is the incorrect size	2	None	10	200	None

Item / Function	Potential Failure Mode(s)	Potential Effect(s) of Failure	Sev	Potential Cause(s)/Mechanism(s) of Failure	Prob	Current Design Controls	Det	RPN	Recommended Action(s)
Delay element	Element burns at an unexpected rate	early/late ignition of the ejection charge	7	Poor mixing from manufacturer or delay spacer is the incorrect size	2	None	10	140	None
Forward Closure	Tearout	Shifting of lining within casing	7	Improper material selection or excessive stress/fatigue or manufacturing defect or stripped threads	2	None	10	140	Run detailed simulations
Forward Closure	Seal is not sufficient	Shifting of lining within casing	4	Improper seating	2	Visual Inspection	1	8	Ensure forward closure is properly attached
Forward Closure	Cracks/Damage	delay element propellant leakage	4	Improper material selection or excessive stress or manufacturing defect	2	Visual Inspection	2	16	Check integrity and replace if needed
Forward Delay Spacer	Incorrect Size	Delay charge could move around, incorrect ignition	3	Assembly defect	2	Visual Inspection	1	6	If team changes delay, ensure spacer reflects the change
Ejection Charge	Failed/improper ignition, incomplete combustion	Parachute deploys at an incorrect time or not at all	10	Poor mixing from manufacturer or Improper spacing from fuse or Improper care of charge (i.e. proper storage, no moisture)	2	None	10	200	Focus on storing grains properly (according to manufacturer specifications)
Ejection Charge	Element burns at an unexpected rate	early/late ignition of the ejection charge	7	Manufacturing defect	1	None	10	70	None
Ejection Charge	Too much charge	More force than expected can damage electronics & recovery system	4	Manufacturing defect	1	None	10	40	None
Ejection Charge	Too little charge	Not enough force to separate the body tubes.	7	Manufacturing defect	1	None	10	70	None
Tracking Smoke	Failed/improper ignition, incomplete combustion	Difficult to locate the rocket	1	Poor mixing from manufacturer or Improper care of charge (i.e. proper storage, no moisture)	2	None	10	20	None
Ejection Cap	Cracks/Damage	Ejection charge could fall out	9	Improper material selection or excessive stress or manufacturing defect	1	Visual Inspection	2	18	Handle rocket with care; regularly check integrity of cap
Ejection Cap	Too hard/brittle	Ejection charge would not properly blast upward	8	Improper material selection or excessive stress or manufacturing defect	1	None	10	80	Check integrity and replace if needed
Ejection Cap	Falls off prior to ignition	Ejection charge falls out or does not light properly	10	Improper seating	2	Visual Inspection	1	20	Ensure ejection cap is properly attached
Motor Retainer	Tearout	Motor could fall out of the body tube or goes up into the rocket	10	Improper material selection or excessive stress/fatigue or manufacturing defect or stripped threads	2	Visual Inspection	5	100	Run detailed simulations
Motor Retainer	Motor is not properly held in	Motor could fall out of the body tube or goes up into the rocket	10	Improper seating	2	Visual Inspection	2	40	Be cautious when installing motor
Motor Retainer	Cracks/Damage	Motor could fall out of the body tube or goes up into the rocket	10	Improper material selection or excessive stress or manufacturing defect	2	Visual Inspection	2	40	Handle rocket with care; regularly check integrity of retainer
Motor Retainer	Retainer threads strip	Motor not seated correctly & can cause instability	8	Excessive stress or manufacturing defect	1	Visual Inspection	1	8	Check integrity and replace if needed
Centering Rings	Glue doesn't bond correctly to airframe	Motor misalignment can cause instability	8	Old/Incorrect glue or assembly defect	3	Visual Inspection	4	96	Check integrity and redo if needed
Centering Rings	Not centered	Motor misalignment can cause instability	8	Assembly/Manufacturing defect	3	Visual Inspection	1	24	Be cautious when installing rings
Fins	Fluttering/Flapping	Rocket is unstable in flight	5	Glue doesn't bond correctly to airframe or opening is too wide or fins are too thin	4	Simulations and visual inspection	3	60	Check integrity and replace if needed; be cautious when gluing

Item / Function	Potential Failure Mode(s)	Potential Effect(s) of Failure	Sev	Potential Cause(s)/Mechanism(s) of Failure	Prob	Current Design Controls	Det	RPN	Recommended Action(s)
Fins	Fins fall out	Rocket is unstable in flight	7	Glue doesn't bond correctly to airframe or opening is too wide or not properly attached	3	None	10	210	Check integrity and replace if needed; be cautious when gluing
Fins	Fins are cracked or damaged	Rocket is unstable in flight	5	Fins are dropped	2	Visual Inspection	2	20	Handle rocket with care
Fins	Fins are misaligned	Rocket is unstable in flight	5	Uneven slots in airframe	4	Visual Inspection	1	20	Check fins regularly after transport
Fin Can	Fluttering/Flapping, unstable flight	Rocket is unstable in flight	5	Glue doesn't bond correctly to motor casing	4	None	10	200	Check integrity and replace if needed; be cautious when gluing
Shock Cord	Cord snaps	Parachute would be disconnected from rocket	10	Manufacturing defect or cord experiences too much force on separation	2	None	10	200	Run detailed simulations
Shock Cord	Cord becomes untied from Eyebolts	Parachute would be disconnected from rocket	10	Assembly defect	4	Visual Inspection	1	40	Be cautious when attaching cord; replace if needed
Shock Cord	Cord is tangled	Parachute may get caught and not deploy	8	Assembly defect	5	Visual Inspection	5	200	Be cautious when assembling rocket for a launch
Shock Cord Eyebolts	Cracked or broken	Shock cord detaches from rocket	10	Manufacturing defect	1	Visual Inspection	1	10	Check integrity and replace if needed
Quicklinks	Cracked or broken	Shock cord detaches from rocket	10	Manufacturing defect	1	Visual Inspection	1	10	Check integrity and replace if needed
Quicklinks	Unhooks from Eyebolts	Shock cord detaches from rocket	10	Assembly defect	2	None	10	200	Be cautious when assembling rocket for a launch
Parachute Canopy	Rips/tears preflight	Rocket does not slow down enough	7	Improper handling, manufacturing defect	1	Visual Inspection	1	7	Check integrity and replace if needed
Parachute Canopy	Rips/tears inflight	Rocket does not slow down enough	7	Improper deployment, getting caught on something, manufacturing defect	2	None	10	140	Be cautious when assembling rocket for a launch
Parachute Canopy	Improper deployment	Uncontrolled descent	10	Improper packing	3	Visual Inspection	3	90	Be cautious when assembling rocket for a launch
Parachute Cords	Tangled	Uncontrolled descent	10	Improper packing, getting caught on something	3	Visual Inspection	4	120	Be cautious when assembling rocket for a launch
Parachute Cords	Snapped	Rocket does not slow down enough	6	Manufacturing defect, deployment at excessive speed	2	None	10	120	Run detailed simulations
Drogue Canopy	Rips/tears preflight	Rocket does not slow down to adequate parachute deployment speed	7	Improper handling, manufacturing defect	1	Visual Inspection	1	7	Check integrity and replace if needed
Drogue Canopy	Rips/tears inflight	Rocket does not slow down to adequate parachute deployment speed	7	Improper deployment, getting caught on something, manufacturing defect, canopy gets burned	2	None	10	140	Be cautious when assembling rocket for a launch
Drogue Canopy	Improper deployment	Rocket does not slow down to adequate parachute deployment speed	6	Improper packing	3	Visual Inspection	3	54	Be cautious when assembling rocket for a launch
Drogue Cords	Tangled	Rocket does not slow down to adequate parachute deployment speed	6	Improper packing, getting caught on something	3	Visual Inspection	4	72	Be cautious when assembling rocket for a launch
Drogue Cords	Snapped	Rocket does not slow down to adequate parachute deployment speed	8	Manufacturing defect, deployment at excessive speed	2	None	10	160	Run detailed simulations
Parachute Blanket	Ripped/torn	Flashes of heat could reach components and set them on fire	9	Manufacturing defect, improper packing	1	Visual Inspection	1	9	Check integrity and replace if needed
Heat Wadding	Amount is insufficient	Flashes of heat could reach components and set them on fire	9	Assembly defect, poor analysis	1	Simulations and visual inspection	3	27	Run detailed simulations

Item / Function	Potential Failure Mode(s)	Potential Effect(s) of Failure	Sev	Potential Cause(s)/Mechanism(s) of Failure	Prob	Current Design Controls	Det	RPN	Recommended Action(s)
Heat Wadding	Wadding does not stop the heat	Flashes of heat could reach components and set them on fire	9	Manufacturing defect	1	Simulations and visual inspection	4	36	Run detailed simulations
Nose Cone	Disconnects from parachute	Nose cone becomes a projectile	10	Manufacturing defect, assembly defect	2	Visual Inspection	5	100	Check integrity of parachute attachment; redo if needed
Nose Cone	Falls off prior to ejection	Rocket loses aerodynamic efficiency and does not go high enough	6	Assembly defect; poor design	3	Visual Inspection	2	36	Ensure fit is good during assembly
Nose Cone	Threads become loose	Payload falls out	6	Manufacturing defect, assembly defect	3	Visual Inspection	2	36	Ensure fit is good during assembly
Nose Cone	Does not release when needed	Nothing gets deployed, no parachute	6	Too Big	3	Visual Inspection	2	36	Ensure fit is good during assembly
Bulkhead	Bulkhead falls out	Payload falls out	6	Epoxy doesn't bond correctly	1	None	10	60	Check integrity and replace if needed; be cautious when gluing
Coupler	Decouples inflight	Rocket becomes two pieces	9	Too small	2	Visual Inspection	3	54	Ensure fit is good during assembly
Coupler	Does not release when needed	Nothing gets deployed, no parachute	10	Too big	3	Visual Inspection	3	90	Ensure fit is good during assembly
Main Rocket Body	Body buckles inflight	Rocket falls apart, yields	10	Manufacturing defect	1	Simulations and visual inspection	1	10	Run detailed simulations
Launch Rail Guides	Rails separate from body	Rocket launches at the wrong angle	8	Screws or epoxy come loose, improper seating, rips through body	1	None	10	80	Check integrity and replace if needed; be cautious when gluing
Payload Release Mechanism	Does not release when needed	Payloads stay within the rocket, payloads launch at too low of an altitude	6	Manufacturing defect, Electrical defect, physical damage	3	Electronics Tests	2	36	Run electronics tests to ensure functionality; replace if needed
Payload Release Mechanism	Releases too early	Rocket interior and chutes gets toasted with a pyro charge	9	Manufacturing defect, Electrical defect, physical damage	3	Electronics Tests	2	54	Run electronics tests to ensure functionality; replace if needed
Drogue Ejection Mechanism	Does not release when needed	Drogue stays within the rocket, drogue deploys at too low of an altitude	7	Manufacturing defect, Electrical defect, physical damage	3	Electronics Tests	2	42	Run electronics tests to ensure functionality; replace if needed
Drogue Ejection Mechanism	Releases too early	Rocket interior and chutes get toasted with a pyro charge	9	Manufacturing defect, Electrical defect, physical damage	3	Electronics Tests	2	54	Run electronics tests to ensure functionality; replace if needed
Parachute Ejection Mechanism	Does not release when needed	Parachute stays within the rocket, parachute deploys at too low of an altitude	9	Manufacturing defect, Electrical defect, physical damage	3	Electronics Tests	2	54	Run electronics tests to ensure functionality; replace if needed
Parachute Ejection Mechanism	Releases too early	Rocket interior and chutes gets toasted with a pyro charge	9	Manufacturing defect, Electrical defect, physical damage	3	Electronics Tests	2	54	Run electronics tests to ensure functionality; replace if needed
Ground Ejection Mechanism (GEM)	Does not release when needed	Payload stays within the rocket	6	Manufacturing defect, Electrical defect, physical damage	3	Electronics Tests	2	36	Run electronics tests to ensure functionality; replace if needed
Ground Ejection Mechanism (GEM)	Releases too early	Payload would eject before landing	5	Manufacturing defect, Electrical defect, physical damage	3	Electronics Tests	2	30	Run electronics tests to ensure functionality; replace if needed
GEM Bearings	Bearings pop off their structure	Payload does not orient correctly for deployment	5	Manufacturing defect, assembly defect, poor tolerancing, physical damage	1	Visual Inspection	1	5	Run detailed simulations
GEM Structure	Structure becomes jammed	Payload does not orient correctly for deployment	5	Manufacturing defect, assembly defect, poor tolerancing, physical damage	4	Visual Inspection	5	100	Run detailed simulations

Item / Function	Potential Failure Mode(s)	Potential Effect(s) of Failure	Sev	Potential Cause(s)/Mechanism(s) of Failure	Prob	Current Design Controls	Det	RPN	Recommended Action(s)
GEM Structure	Breaks	Payload does not orient correctly for deployment	5	Manufacturing defect, assembly defect, poor tolerancing, physical damage	2	None	10	100	Run detailed simulations
GEM Stability System	Weight dislodges from structure	Payload does not orient correctly for deployment	5	Manufacturing defect, assembly defect, poor tolerancing, physical damage	2	Visual Inspection	5	50	Run detailed simulations
Battery	Improper output voltage	damage to components, components do not turn on	10	traces not labeled or connected to proper terminals, choice of battery	7	use LED's as a peripheral to show if proper voltage is being supplied to the PCB	1	70	Use a new battery,
Battery	improper current supplied to system	no current sent to components, damage to components	10	Manufactured defect, traces not labeled or connected to proper terminals	3	calculating the current draw of peripherals and following datasheets	2	60	Use a new battery, change peripheral models
MCU	Programing errors	desired commands are not initiated on the board and corrupted data transfer occurs	10	invalid trace length, not paying attention to orientation labels, high density IC, Orientation	6	Running code simulations	5	300	Test with development board, check wiring and chip, replace and resolder if necessary
MCU	Watchdog resets	No commands are initiated on the board and no data transfer occurs, coding errors,	10	overloading data with too much information due to high sampling rate on multiple sensors	2	selecting sampling rates by importance of information	1	20	changing the sampling rate if this persists
MCU	damaged	No commands are initiated on the board and no data transfer occurs	10	too much voltage supplied, physically damaged,	2	double checking Kicad labels for the pins and we can also use a silk screen to write the texts on the PCB to confirm after printing	1	20	Test with development board, check wiring and chip, replace and resolder if necessary
Regulator	Desired voltage isn't reached	supplying too much power or not enough power to components leading to either overheating/damage	10	amount of supplied voltage, model of regulator, compatibility to chosen MCU	9	use LED's as a peripheral to show if proper voltage is being supplied to the PCB	1	90	Redesign regulator if original is wrong design; replace components to double-check
Regulator	Improper output current	can't support current draw of peripherals	10	not choosing correct models for regulator for desired current output	2	researching models that support current requirements of peripherals	1	20	Redesign regulator if original is wrong design; replace components to double-check
IMU	Doesn't measure or transmit flight data to MCU	no flight data to be sent to ground station	7	improper wire connections, improperly built PCB	5	use LED's as a peripheral to show if proper voltage is being supplied to the PCB Test PCB on breadboard for measurements, label pins using silkscreen.	6	210	Test with development board, check wiring and chip, replace and resolder if necessary
IMU	inaccurate readings	inaccurate readings sent to ground station	1	acceleration of rocket, low transmission rate	7	using models that can withstand high speeds and have high accuracy	4	28	replace IMU model
Altimeter	Doesn't measure or transmit flight data to MCU	actuators never activated, payloads never get released from rocket, no flight data sent to ground station	10	improper wire connections, improperly built PCB	5	Use LED to test voltage through device. Test PCB on breadboard for measurements, label pins using silkscreen	1	50	Test with development board, check wiring and chip, replace and resolder if necessary
Altimeter	inaccurate readings	inaccurate readings sent to ground station, payloads/parachutes don't release at desired altitudes	10	acceleration of rocket, low transmission rate, rapid pressure change, out of sensing range	5	using models that can withstand high speeds and have high accuracy	1	50	replace altimeter

Item / Function	Potential Failure Mode(s)	Potential Effect(s) of Failure	Sev	Potential Cause(s)/Mechanism(s) of Failure	Prob	Current Design Controls	Det	RPN	Recommended Action(s)
Transceiver	does not convert data from MCU into analog	no data received on ground station, rocket can not be recovered, data corrupted	7	having the transceiver too close to other components, model of transceiver used, wiring and connections	3	Use a conversion component onto board to transfer to google maps	1	21	Test
Transceiver	does not send data to ground station	no data received on ground station, rocket can not be recovered, data corrupted	7	antenna isn't supplying desired gain, groundstation out of range model of transceiver	8	use an active antenna to increase gain and cover wider distance	1	56	Test
GPS	Failure to detect satellites,	Low tracking accuracy or failure to track,	8	model used, signal strength	3	using a module that satisfies requirements	4	96	Test
GPS	GPS locks during ascent	current location of rocket is not sent to ground station only position before lock,	7	model used, acceleration of rocket	1	choosing a module that supports high speeds	1	7	replace module
Actuator	Desired motion not reached	springs aren't released when desired altitude reached, door doesn't open to release payloads from body tube	10	not enough voltage supplied to power actuator, draws too much current from the battery/regulator	10	Measure voltage and current when testing actuator movement	2	200	Replace actuator
Beeper/Buzzer	Doesn't go off	Rocket/Payloads harder to find, possibly never recovered	8	no power supplied to beeper, damaged component	2	proper wiring	2	32	Test
Beeper/Buzzer	Distance the sound travels	Rocket/Payloads harder to find, possibly never recovered	8	speakers covered/blocked by environment	7	Model used for frequency and range	1	56	Noise range test
Antenna	doesn't transmit at a frequency that covers 5k+ feet	data wont get picked up by ground station, difficult to recover rocket, cant read flat data	10	bent antenna, too close to other components leading to noise and corruption, disconnecting from PCB	8	Use commonly used models,	1	80	Range Test
Surface Mount Devices Components	Doesn't fit traces on PCB	not being able to see components, can't solder components, can't properly control current flowing through system	10	using components (capacitor, resistor, inductor) smaller than an 805, ordering the wrong parts	2	Measure each component, correct size during PCB design	1	20	Redesign PCB to accommodate correct size
Male/Female headers	Falls out during flight	peripherals aren't powered by main power supply, data isn't transferred to the MCU from peripherals	10	not properly soldered, vibration, improper orientation on PCB, Improper pin labels on Kicad	2	reviewing PCB labels and checking names and connections and printing with a silkscreen when soldering connections	4	80	Check wiring, replace and resolder if necessary
Male/Female headers	not enough surface area on board to support all headers	not being able to house necessary peripherals	3	improper wire connections, improperly built PCB	4	using wires to connect peripherals if necessary	1	12	Check size on PCB and headers to see if they match
Wire Connection	no voltage and current transferred to peripherals	no power delivered to components and no functions occur	10	Connections fall out during flight, wire breaks, improper wiring	2	keeping boards attached using male and female headers to avoid using wires	1	20	Check connections, measure voltage and current
SD card	Failure to store data	not being able to review all/any flight data	5	lack of storage space, data never sent due to improper wiring	3	labeling wires and choosing SD card with ample storage	10	150	check wiring, leave labels, choose different SD card
Electronic Bay	Moisture leaks	Electronics could short	10	Cracks or leaks in the rocket, exposing the electronic bay	2	Using airtight structures	7	140	Check holes and use sealant
Electronic Bay	Bad mounting	Components are loosen and can damage other parts	10	Screws not secured to board, bay not contained to rocket	2	Using all corners on PCB for screw holes	3	60	Using long screws and lock nuts
Overall Electronics system	Overheating	all components stop functioning	10	too much voltage supplied, poor regulation, improper connections, lack of insulation	2	checking connections, using regulators that can supply desired voltage, proper spacing	5	100	Test
USB	Doesn't match port on device	data isn't sent to the terminal to display flight data	6	not choosing proper component when designing ground station PCB	1	Using a MCU module that supports a USB connection	1	6	Test
External Logging Device	Doesn't properly show flight data	terminal does not display the data and location is not displayed on google maps	7	Ground-station software cannot decode telemetry packets due to packet loss, baud mismatch, or improper data formatting	2	Running code simulations to make sure data is being converted from the MCU to terminal before launch	1	14	Test

Item / Function	Potential Failure Mode(s)	Potential Effect(s) of Failure	Sev	Potential Cause(s)/Mechanism(s) of Failure	Prob	Current Design Controls	Det	RPN	Recommended Action(s)
External Logging Device	no battery	terminal does not display the data and location is not displayed on google maps	7	not charging laptop or phone	1	making sure everyone's devices are charged	1	7	charging before demo
External Logging Device	no device present	terminal does not display the data and location is not displayed on google maps	7	no one brings s a device that will display flight data	1	having at least 2 people confirm that they have their devices present	1	7	bring a device
PCB's	manufacturer does not properly sketch wire tracings	power isn't distributed properly, boards don't perform intended goals	10	issues at the shop, wire tracings too thin and missed	2	having a set trace length for wires, avoiding small components	5	100	manually sketch wires in SD2 lab
PCB's	stencil doesnt match pcb outline	difficulty soldering parts, components not aligned system fails	2	not adding the proper measurements when ordering PCBs for stencil lengths, not ordering the stencils at all	3	confirming the dimensions of the stencils when making orders	10	60	solder without the stencil and be careful
PCB's	burn through wire connection while soldering	no information or power sent to a/multiple components, intend results not met	7	using too much heat when soldering, holding the pen to the PCB for too long	1	following SD recommendations for soldering temperature and being safe	1	7	solder on new wire to recreate the connection
PCB's	not having enough space for peripherals	not having all peripherals integrated onto main PCB using male/female headers	7	not taking measurements of peripherals into account when adding the headers onto the main PCB	3	using the measure tool on the silk screen to outline peripheral dimensions	2	42	stacking headers or having peripherals at an angle

Table 13. Failure Mode and Effects Analysis

A Failure Modes and Effects Analysis (FMEA) was conducted to identify the critical areas of concern for the selected concept. For each component of the design, failure modes were identified to represent the various things that may cause a part to be unable to perform its required function. The team listed potential causes of each failure mode to serve as a base for determining ways to mitigate the risk each failure mode presents. To determine which failure modes were most significant, the potential effects of each were analyzed and given a severity (Sev) score out of ten. Additionally, the probability (Prob) and detectability (Det) of each failure mode were given a similar ranking.

When evaluating severity, a basic framework was laid out to guide the team's ratings: a score of 1 represented negligible damage or minor costs as a result of the failure mode; a score of 5 represented the potential for medium damage or a minor risk of injury; and a score of 10 represented the possibility of a major injury, major cost, or a fatality. Additionally, from the context of this project as part of a capstone class, failure modes that might result in critical

requirements not being met (such as the safe deployment of payloads) were assigned severity scores of up to 7. However, the severity scores of 8 through 10 were reserved for effects that can lead to significant bodily or property harm outside of the scope of the project to ensure that safety is always paramount when making engineering decisions.

The probability score was determined on an approximately linear scale. A score of 10 roughly represents a 10% chance of the failure mode occurring, and a score of 1 represents a case that almost never happens and is a negligible concern. Probability scores of 1 were assigned very sparingly, typically for failure modes caused by manufacturing defects of simple parts. The detectability score shows how likely the team is to find the causes of the failure and correct them before the failure occurs. Higher detectability scores are, counterintuitively, less detectable failure modes. A 10 in this category would be a failure mode that is impossible to detect before it fails, and a 1 is a failure mode that can be detected very easily and with minimal testing.

All of the severity, probability, and detectability scores were combined into a Risk Priority Number to compare the significance of each failure mode. Risk Priority Numbers (RPN) were calculated by simply multiplying the severity, probability, and detectability.

$$RPN = Sev * Prob * Det \quad (1)$$

A significant example is the motor's ejection charge not firing properly (RPN of 200), leading to the parachute not deploying properly, and possibly, uncontrolled descent. An uncontrolled descent is automatically assigned a severity score of 10, as it can cause significant property damage or death. Additionally, the detectability score was deemed a 10, as the cause of this failure mode is poor-quality grains, something that will not be discoverable without advanced testing and destruction of said grains. However, the probability of this occurring is low

so long as the team properly stores the motor and its components. Thus, probability was given a 2, and the failure mode was deemed significant but preventable.

Another example is the main parachute chords getting tangled (RPN of 120). If the parachute fails to inflate, the rocket can again go into an uncontrolled descent, and so the severity is, again, a 10. This time, however, the failure mode is detectable by checking the parachute packing before launch, leading to a detectability score of 4 to account for human error. The probability is also low and was given a 3, leading this failure mode to be less significant overall than the ejection charge as long as the group performs its preflight checks properly.

This highlights the purpose behind an FMEA. By this method, critical sources of risk can be dissected and thoroughly accounted for to give the project its highest chance of success. By comparing each of the modes and their RPN, the team can determine which risks need to be specially accounted for in the design stage, with redundancy if necessary, and which risks can be mitigated by basic checks.

X. Significant Accomplishments and Future Work

To date, PLEIADES has built a strong foundation for the Rocket Payload Deployment System through extensive development and research. There are weekly meetings to discuss milestone goals and tasks. Each milestone contributed to development, from initial research and definition requirements to concept generation and evaluation. In September, the process began with M1. M1 consisted of the technology memo study in which the team allocated members to research various technical areas related to the project. It served as guidance to become familiar with past work and information relevant to the project. The shared collection of M1 memos

provided the team with a common vocabulary and foundational understanding that supported later engineering decisions.

Built on the foundation of M1, the first few weeks of October were used to work on M2, beginning with functional and component decompositions. The functional and component decompositions were discussed and created, enabling the team to identify all critical subsystems, including avionics, deployment mechanisms, telemetry, and environmental monitoring. The team generated a broad set of conceptual solutions for each major subsystem and illustrated these within a comprehensive concept map, showing the relationship between electronics, mechanics, and goals. converted stakeholder goals and safety rules into a structured set of customer requirements, engineering requirements, and a matrix. The house of quality (HoQ) was completed using updated customer and technical specifications, clarifying which design metrics, such as deployment accuracy, timing reliability, payload safety, and rocket flight, carry the greatest relative weight. The customer and technical requirements were correlated against one another to evaluate their relationship and strength. The technical requirements were weighed against one another to determine their correlation. The engineering requirements were given a target value/tolerance and a validation method.

With requirements formally defined, the team progressed to M3, a concept development report. The remaining time for senior design 1 is used for M3, research, and development. The team began by creating multiple Pugh matrices to compare alternative concepts for propulsion, deployment mechanism, electronics, and telemetry. These analyses led to the identification of weighted design directions and eliminated several concepts with excessive complexity or lower reliability. The team also completed a preliminary FMEA for the mechanical, aerospace, and electrical aspects. It highlighted several high-risk areas, such as mechanical jamming,

insufficient actuation force, environmental sensitivity of electronics, and power interruption during flight. These findings are already informing design priorities for prototyping and testing.

Future work will include Statement of Work and Project Plan (M4), Detail System Design Review and Report (M5), and Complete Design Documentation Package (M6). The team plans to spend the time after M3 simulating, modeling, and designing the rocket and its payload system. The main research should be concluded by the end of the semester. Modeling and designing the system should be finished by the end of winter break. Ideally, all parts would be ready by the beginning of senior design 2. The tests include electronics, payload mechanisms, stress on the frame, parachute deployment, data transmission, separation test, etc. The period of testing will ideally be done before March 1st, as the 14th is the next available date for the Palm Bay rocket launch. The latest launch date is April 11th.

XI. Conclusions and Recommendations

Based on the project's efforts to date, several conclusions emerged regarding the rocket design and deployment method. The HoQ established that the most critical performance factors were the rocket itself and the deployment. The concept generation and Pugh matrices helped formalize different ideas for propulsion and electronics, and compared them for evaluation. The trends suggest that reliability and testability should take precedence over mechanical complexity. The FMEA revealed that the highest risks were associated with mechanical and electronic components. These risks must be mitigated early through prototype testing and should drive it towards a robust system with fixable areas. Ensuring that these parts are optimal will minimize issues in the manufacturing process.

Recommendations moving forward:

- Prioritize early prototyping of the deployment mechanism to identify jamming, friction, and tolerance issues before committing to actual implementation.
- Begin electronics early, focusing on MCU selection, IMU/altimeter integration, actuator conditioning, and telemetry.
- Validate environmental requirements through simulations and testing to ensure that the components meet the specified conditions.
- Refine the system by developing preliminary schematic diagrams, CAD layouts, and payload housing.
- Conduct early integration testing between mechanical actuation and avionics to confirm timing accuracy and deployment reliability.

Together, these conclusions and recommendations provide a clear path forward for the next phase of the project, including detailed design, prototyping, and testing.

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Appendices

Appendix A: ABET Design Matrix

	Critical / Main Contributor	Strong Contributor	Necessary but not a Primary Contributor	Necessary but only a Minor Contributor	Only a Passing Reference	Not Included in this Project	Report Section
AERONAUTICAL							
Aerodynamics	X						VI
Aerospace Materials			X				V
Flight Mechanics	X						VIII
Propulsion		X					V
Stability & Control		X					V
Structures			X				V
ASTRONAUTICAL							
Aerospace Materials			X				V
Attitude Determination & Control		X					VI

Orbital Mechanics			X				III
Rocket Propulsion		X					V
Space Environment				X			III
Space Structures				X			III
Telecommunications		X					V
MECHANICAL							
Thermal-Fluid Energy Systems						X	-
Machines & Mechanical Systems			X				V
Controls & Mechatronics			X				VI
Materials Selection	X						V
Modeling & Measurement Systems		X					VIII
Manufacturing		X					VIII
ELECTRICAL & COMPUTER							
Circuit Analysis	X						VIII
Embedded Systems / Microcontrollers	X						V
PCB Design & Layout	X						III
Control System		X					VI
Power Electronics & Regulation		X					III
Signal Processing		X					III
Communication Systems	X						V

Table 14. ABET Matrices

Appendix B: Concept Generation

A. User Needs & Engineering Requirements

Customer Requirements (what's?)	Relative Wt.
Safe Flight of Rocket	0.106
Carry Payloads	0.092
Affordable	0.085
Safe deployment of Payloads	0.083
Aerodynamic Performance	0.081
Easy Transportation	0.076
Recoverable Rocket	0.075
Location Tracking Accuracy	0.072
In-flight Data Acquisition	0.066
Recoverable Payloads	0.053
Easy Payload Loading	0.051
Altitude Tracking Accuracy	0.045
Store Data	0.036
Time Tracking Accuracy	0.035
Attitude Tracking Accuracy	0.026
Speed Tracking Accuracy	0.018

Table 15. User Needs

Reference Number	Requirement	Target Value / Tolerances	Validation Method
C.I.1	Motor should be powerful enough to conduct the launch	320-640Ns	Inspection
C.VII	Rocket must be compatible with the launch system	Has launch rail guides	Inspection
C.VIII	Battery must power the electronics for the entire duration of the launch	7.5Wh min	Test
C.XI	Must carry a set amount of payloads	3	Inspection
C.XVI	Storage must be large enough to log all flight data	10gB min	Inspection
C.XVI.1	Location uncertainty must be within a certain tolerance	10m max	Analysis
C.XVI.1	Altitude uncertainty must be within a certain tolerance	10m max	Analysis
C.XVI.1	Time uncertainty must be within a certain tolerance	1s max	Analysis
C.XVI.1	Attitude uncertainty must be within a certain tolerance	15 deg max	Analysis

C.XVI.1	Speed uncertainty must be within a certain tolerance	20m/s max	Analysis
C.XVII.1	Rocket should transfer data quick enough to get real-time measurements	TBD	Test
C.XXIV	Payload should slow on descent to a safe landing speed	11m/s max	Test
F.I.1	Rocket must reach a minimum ascent speed	20m/s min	Test
F.I.2	Must reach a minimum altitude	1320 ft min	Test
F.II	Drag Coefficient of the rocket body on launch should be under a set maximum	0.8 max	Analysis
F.III	Rocket must be launched in suitable weather conditions	Must not launch directly in clouds.	Inspection
F.IV	Motor shall not be ignited by use of a mercury switch or roller switch		Inspection
F.VI	Rocket should be easy to assemble	30min max	Test
F.VII.1	Stability Margin must be within a set range	$1.5-2 \pm 0.2$	Analysis
F.XI	Rocket must be modular	2 sections minimum	Inspection
F.XII	Rocket should be horizontal after landing	$(0,0,0) \pm (180,10,10)$	Test
F.XII.1	Rocket must slow on descent to a safe landing speed	11m/s max	Test
F.XIII	Electronics must have a sampling rate quick enough to get accurate data	$100\text{Hz} \pm 10\text{Hz}$	Test
F.XIV.1	Payloads should be loaded within a set time	1m max for each payload	Test
F.XV	Deployments must be within a maximum time duration	10s max	Test
F.XV.1	Rocket should be oriented properly when deploying payloads	$(0,90,0) \pm (180,20,20)$	Test
F.XV.1	Deployment should create an attitude shock within a reasonable limit	15 deg max	Test
F.XV.4	Payload deployments should be sequential with a minimum time spacing	3s min	Test
F.XVI.1	Rocket must be large enough to house all components	Dependent on Motor/Payload Sizing	Inspection
F.XX	Rocket must be produced with available machines	Tolerancing: TBD	Inspection
F.XXI	Rocket's factor of safety must be within a set range	$2-4 \pm 0.5$	Analysis
F.XXI	The payload deployment's factor of safety must be within a set range	$2-4 \pm 0.5$	Analysis
F.XXII	Rocket must withstand dynamic pressure	$6\text{kPa} \pm 1\text{kPa}$	Test
H.III	Costs should stay below a set target	\$700 max	Inspection

Table 16. Engineering Requirements

B. Relationship Matrix of User Needs to Engineering Requirements

Quality Function Deployment

Project Title: Rocket with Deployable Payloads
Project Leader: _____
Date: _____

Customer Requirements (what's?)	Relative Wt. (0.00-1.00)	Number of Payloads	Cost (\$)	Altitude (ft)	Deployment Time after Launch (s)	Deployment Altitude (ft)	Dimensions (m) ³	Rocket Descent Speed (m/s)	Sampling Rate (Hz)	Location Uncertainty (km)	Altitude Uncertainty (km)	Time Uncertainty (s)	Speed Uncertainty (km/s)	Transfer Error (m)	Deployment Altitude Error (km)	Deployment Duration (s)	Battery Life (Wh)	Payload Descent Speed (m/s)	Payload Loading Time (s)	Data Storage Capacity (mb)	Ascent Speed (m/s)	Acceleration (m/s ²)	Drag Coefficient	Lift Coefficient	Stability Margin (CA)	Rocket Factor of Safety (FoS)	Factor of Safety (FoS) - Payload
Safe Flight of Rocket	0.106	-5	3	-1			1	-9							-3	1					3	3	-3	-1	9	9	
Carry Payloads	0.092	9	1	1			9									-3			1		-1						
Affordable	0.085	1	-9				-9	-3	-3								-3	-3		-3	-3						
Safe deployment of Payloads	0.083	3	3	9	9	9	-9			-3	-3	-3	-3	-9	9						-9	-3			1	3	
Aerodynamic Performance	0.081	-1	3				3														9	9	9	9	3		
Easy Transportation	0.076	1					-9																				
Recoverable Rocket	0.075	1	-9				1	-9														-3			9		
Location Tracking Accuracy	0.072	1					-1	9	-9								9				-3	-3					
In-flight Data Acquisition	0.066	1	-1				9							9			9				-9						
Recoverable Payloads	0.053	-3	1	-9	9													-9									
Easy Payload Loading	0.051	-1	1			3													-9		-1						
Altitude Tracking Accuracy	0.045	1					9	-9									9				-3	-3					
Store Data	0.036	1	-1			-1	-3						9				9			9							
Time Tracking Accuracy	0.035	1					9		-9								9				-3	-3					
Attitude Tracking Accuracy	0.026	1					9		-9								9				-3	-3					
Speed Tracking Accuracy	0.018	1					9					-9					9				-3	-3					

Table 17. House of Quality

C. Relevant Engineering Standards

- PCB Standards
 - IPC-2221
- Communication Standards
 - I2C

- SPI
- UART
- IEC 61800
- LoRaWAN v1.0.4 Specification
- FCC Part 15

D. System Concept Alternatives

Propulsion Pugh Matrix					
		Solution Alternatives			
		Importance Rating	Single Motor	Motor Cluster	Two Stage Motor
<u>Concept Selection Legend</u>					
Better	+				
Same	S				
Worse	-				
Key Criteria					
Cost	7				
Weight	2				
Ease of Assmbly	3				
Complexity of Ignition	8				
Reliability	5				
Safety to Rocket	2				
Safety to Surroundings	10				
Portability	2				
Sum of Positives			0	1	
Sum of Negatives			5	6	
Sum of Sames			3	1	
Weighted Sum of Positives			0	10	
Weighted Sum of Negatives			22	27	
TOTALS			-22	-17	

Table 18. Propulsion Configuration Pugh Matrix

In-flight Payload Deployment Pugh Matrix													
			Solution Alternatives										
			Importance Rating	Spring Ejection - Top	Spring Ejection - Side	Spring Ejection - Bottom	CO2 Ejection - Top	CO2 Ejection - Side	CO2 Ejection - Bottom	Pyro Ejection - Top	Pyro Ejection - Side	Pyro Ejection - Bottom	Drogue Ejection - Top
Key Criteria													
Cost			7	-	S	-	-	-	-	-	-	+	+
Weight			3	-	S	-	-	-	+	+	+	+	+
Ease of Assembly			5	-	S	-	-	-	+	+	+	+	+
Deployment Reliability			10	-	S	+	S	+	+	+	+	S	-
Safety to Payload			4	S	S	S	S	S	-	-	-	S	S
Testing Ease			3	S	S	-	-	-	-	-	-	-	+
Sum of Positives				0	0	1	0	1	3	3	3	3	4
Sum of Negatives				4	0	4	4	4	3	3	3	1	1
Sum of Sames				2	6	1	2	1	0	0	0	2	1
Weighted Sum of Positives				0	0	10	0	10	18	18	18	15	18
Weighted Sum of Negatives				25	0	18	18	18	14	14	14	3	10
TOTALS				-25	0	-8	-18	-8	4	4	4	12	8

Table 19. In-flight Payload Deployment Pugh Matrix

Ground Payload Deployment Pugh Matrix												
			Solution Alternatives									
			Importance Rating	Spring - Top - Passive Gyroscope	Arm - Side - Autonomous	Arm - Side - Remote Controlled	Spring - Top - Active Gyroscope	Spring - Top - Parachute Insured	Spring - Top - External Body Rotation	Spring - Side - Passive Gyroscope	Spring - Side - Active Gyroscope	Spring - Side - Parachute Insured
Key Criteria												
Cost			7	-	-	-	S	-	S	-	S	-
Weight			3	-	-	-	+	-	S	-	S	-
Ease of Assembly			5	-	-	-	+	-	-	-	+	-
Deployment Reliability			10	+	+	-	-	-	S	-	-	-
Safety to Payload			4	+	+	S	S	S	S	S	S	S
Testing Ease			3	-	-	S	-	S	S	S	-	S
Complexity			5	-	-	-	S	-	S	-	S	-
Sum of Positives				2	2	0	2	0	0	0	1	0
Sum of Negatives				5	5	5	2	5	1	5	2	5
Sum of Sames				0	0	2	3	2	6	2	4	2
Weighted Sum of Positives				14	14	0	8	0	0	0	5	0
Weighted Sum of Negatives				23	23	30	13	30	5	30	13	30
TOTALS				-9	-9	-30	-5	-30	-5	-30	-8	-30

Table 20. Ground Payload Deployment Pugh Matrix

Inertial Measurement Unit Pugh Matrix						
		Solution Alternatives				
Key Criteria	Importance Rating	Benchmark Option (GY-521 MPU6050)	SparkFun (ICM-20948)	Adafruit TDK InvenSense (ICM-20948)	Adafruit 9-DOF Absolute (BNO055)	SparkFun (LSM9DS1)
	Concept Selection Legend Better + Same S Worse -					
	Supply Voltage	8	S	S	S	S
	Current	10	+	+	-	-
	Number of interfaces	2	+	+	S	S
	Degrees of Freedom	2	+	+	+	+
	Cost	4	-	-	-	-
	Temperature Range	3	-	-	-	-
	Full-scale ranges of sensors	5	+	+	+	+
	Size	5	-	S	-	+
Frequency	8	S	S	S	S	
Sum of Positives			4	4	2	3
Sum of Negatives			3	2	4	3
Sum of Sames			2	3	3	3
Weighted Sum of Positives			19	19	7	12
Weighted Sum of Negatives			12	7	22	17
TOTALS			7	12	-15	-5

Table 21. Inertial Measurement Unit Pugh Matrix

Global Positioning System Pugh Matrix					
		Solution Alternatives			
Key Criteria	Importance Rating	MAX-M10S	NEO-M9N	MT3339	ZED-F9P
	Precision	10	-	-	+
	Cost	6	S	+	-
	Size	8	-	-	-
	Navigation sensitivity	7	S	S	S
	Update rate	4	+	-	S
	power consumption	7	-	-	-
	Supply voltage	3	S	S	S
	interfaces	3	+	+	+
	Sum of Positives		2	2	2
Sum of Negatives		3	4	3	
Sum of Sames		3	2	3	
Weighted Sum of Positives		7	9	13	
Weighted Sum of Negatives		25	29	21	
TOTALS		-18	-20	-8	

Table 22. Global Positioning System Pugh Matrix

Onboard Battery Pugh Matrix

<u>Concept Selection Legend</u> Better + Same S Worse -		Solution Alternatives			
		Importance Rating	Alkaline	Lithium-Ion	Lithium-Polymer
Key Criteria					
cost	6		-	-	-
nominal voltage	4		+	+	-
capacity	10		+	S	+
lifetime	7		+	+	+
size	8		S	S	-
Sum of Positives			3	2	2
Sum of Negatives			1	1	3
Sum of Sames			1	2	0
Weighted Sum of Positives			21	11	17
Weighted Sum of Negatives			6	6	18
TOTALS			15	5	-1

Table 23. Onboard Battery Pugh Matrix

Location Beeper Pugh Matrix					
		Solution Alternatives			
		Importance Rating	Passive piezo buzzer	Active buzzer	Electronic beeper module
Concept Selection Legend Better + Same S Worse -					
Key Criteria					
Sound level	10		+	+	
Power consumption	3		-	-	
Control	2		+	+	
Size	8		-	-	
Cost	8		-	-	
Sum of Positives			2	2	
Sum of Negatives			3	3	
Sum of Sames			0	0	
Weighted Sum of Positives			12	12	
Weighted Sum of Negatives			19	19	
TOTALS			-7	-7	

Table 24. Location Beeper Pugh Matrix

Telemetry Antenna Pugh Matrix					
		Solution Alternatives			
Key Criteria	Importance Rating	Active	Passive	Patch	PCB
	typical gain	10	-	-	-
	power requirement	2	+	+	+
	size	6	-	+	+
	cost	5	+	+	+
	suitability for long distance	10	-	-	-
	Sum of Positives		2	3	3
Sum of Negatives		3	2	2	
Sum of Sames		0	0	0	
Weighted Sum of Positives		7	13	13	
Weighted Sum of Negatives		26	20	20	
TOTALS			-19	-7	-7

Concept Selection Legend

Better +

Same S

Worse -

Table 25. Telemetry Antenna Pugh Matrix

Actuator Types Pugh Matrix						
		Solution Alternatives				

Table 26. Actuator Types Pugh Matrix

Actuator Models Pugh Matrix										
			Solution Alternatives							
Concept Selection Legend Better + Same S Worse -			Importance Rating	Benchmark Option (SG90)	MG996R	HS-422	Mini Push-Pull Solenoid 2776	Large push-pull solenoid 413	PQ12-30-6-R	PA-12-10613474T
Key Criteria										
Voltage	9			S	S	S	-	S	-	
Current	9			-	-	-	-	-	-	
Torque				+	+					
Operating Speed				-	-			-	-	
Degree of Motion				S	+					
Weight				-	-	-	-	-	-	
Cost				-	-	-	-	-	-	
Size				-	-	-	-	-	-	
Sum of Positives				1	2	0	0	0	0	
Sum of Negatives				5	5	4	5	5	6	
Sum of Sames				2	1	1	0	1	0	
Weighted Sum of Positives				0	0	0	0	0	0	
Weighted Sum of Negatives				9	9	9	18	9	18	
TOTALS				-9	-9	-9	-18	-9	-18	

Table 27. Actuator Models Pugh Matrix

E. Decompositions

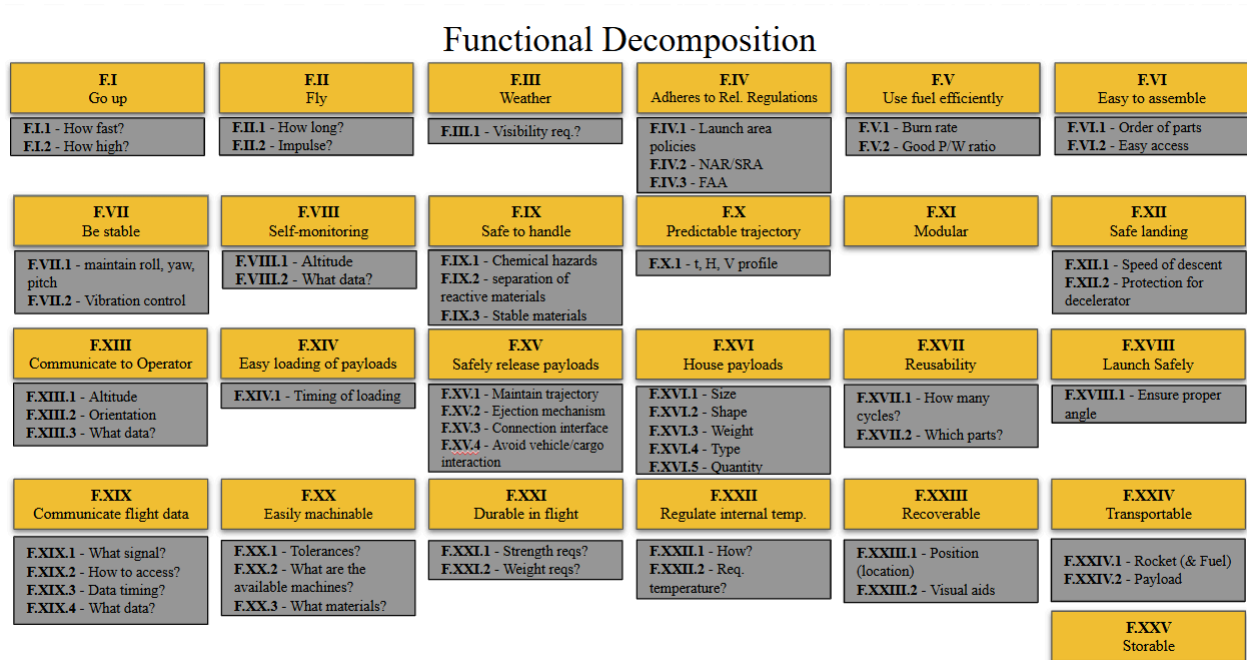


Figure 12. Functional Decomposition

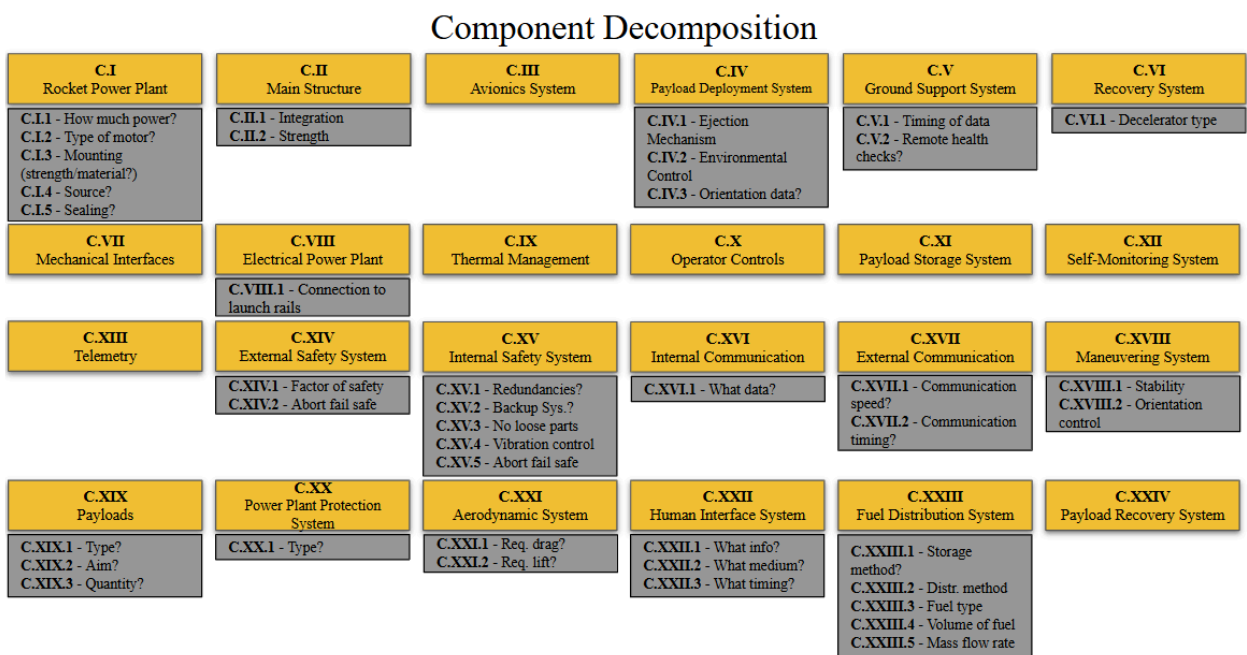


Figure 13. Component Decomposition